

A Hybrid Corpus-to-Taxonomy Pipeline for Physical AI Risk Cards

Data Collection, Processing, Deduplication, and Hierarchical Assignment

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Executive Summary

This manuscript documents the data collection, processing, analysis, and risk-card generation procedure used to construct the Physical AI slice of the Responsible AI Risk Taxonomy v2.0. The workflow follows a hybrid taxonomy design: high-level structure is anchored in authoritative safety, risk-management, and embodied-AI references, while L4 risk cards are generated bottom-up from a multi-source corpus of scientific publications, policy reports, and patents. The updated local integrated document space contains 3,906,767 AI records, 741,428 AI infrastructure records, 383,149 Physical AI records, and 82,971 Physical AI risk records after within-group deduplication across scientific publications, policy reports, and patents. The released local taxonomy snapshot contains 182 L4 risk cards, 24 L3 sub-categories, 3 L2 categories, and 360 card-level evidence/reference rows. Two core procedures are formalized: (i) risk candidate generation and semantic deduplication, and (ii) EM-based L3 family assignment.

Keywords: Physical AI; AI risk taxonomy; robotics safety; embodied AI; cyber-physical systems; semantic deduplication; BGE-M3; HDBSCAN; reproducible technical report.

Reproducibility statement. This report is written as a technical reconstruction protocol. It specifies the source families, retrieval artifacts, local file denominators, filtering thresholds, embedding inputs, clustering and deduplication procedures, hierarchy assignment procedure, validation checks, and release artifacts required to reproduce the Physical AI Risk Taxonomy construction workflow.

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1 Scope and Design Rationale

Physical AI is defined here as AI deployed in systems that sense, reason, and act in the physical world, including robots, humanoids, drones, autonomous vehicles, cyber-physical systems, and embodied agents. This scope follows recent Physical AI and embodied-safety work that treats world models, robot constitutions, and embodied benchmarks as distinct from text-only AI risk settings [15, 22, 23]. The taxonomy is organized as:

L0 Ownership Levels → *L1* Physical AI Risks
→ *L2* Categories → *L3* Sub-categories → *L4* Risk Cards.

The *L2* layer distinguishes the locus of risk responsibility: internal system safety, interaction safety at the human/environment interface, and broader societal safety. The *L4* layer is the operational unit: each card has a stable identifier, bilingual label and definition, severity and probability proxies, short evidence justifications, references, and 3H1R alignment tags.

The methodological problem is that Physical AI risk taxonomies cannot be built purely top-down or purely bottom-up. A top-down taxonomy is stable and institutionally legible, but may miss emerging embodied, agentic, and cyber-physical failure modes [33, 32]. A bottom-up corpus pipeline captures emerging risks, but can produce noisy, overlapping, or non-actionable categories. The adopted design therefore combines: (1) theory-anchored upper levels, (2) corpus-driven *L4* candidate discovery, (3) semantic deduplication, and (4) expert review for boundary cases. This hybrid stance is consistent with prior language-model risk taxonomies and AI risk repository work, which combine expert structure with systematic evidence aggregation [31, 32].

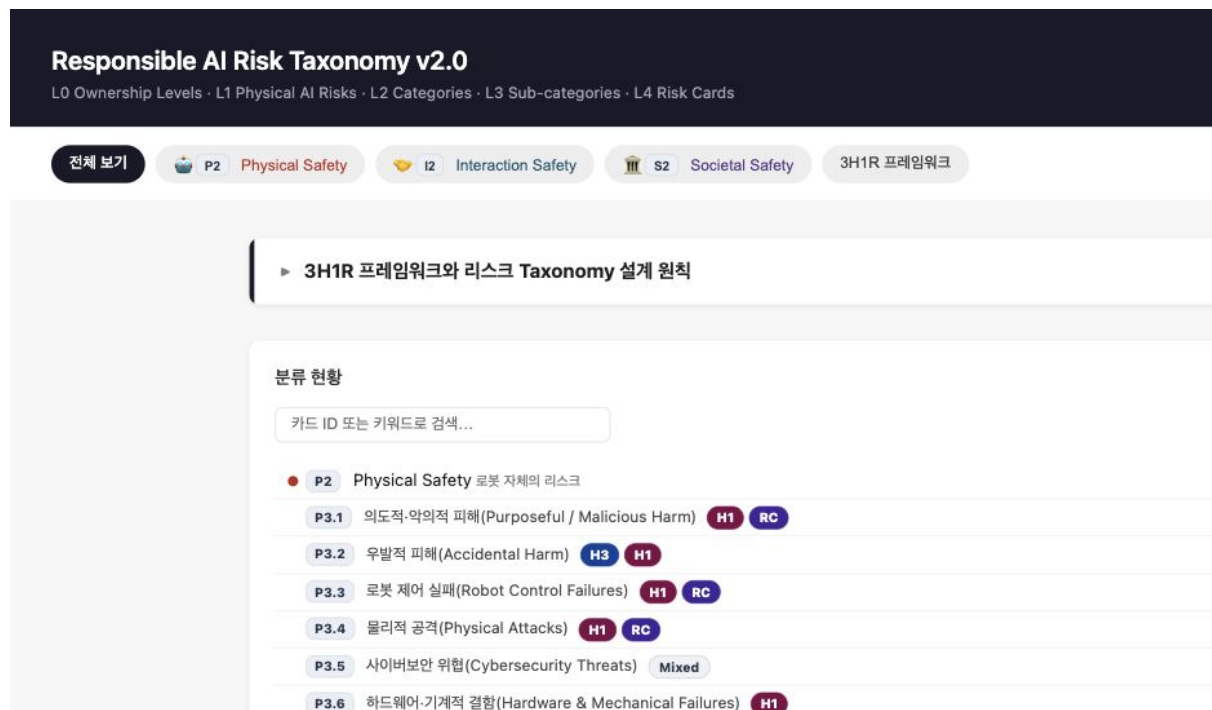


Figure 1: Released taxonomy interface

Note: The interface is used as the visual inspection layer for the Physical AI Risk Taxonomy. It presents the *L0*–*L4* hierarchy, *L2* source categories, *L3* sub-category distribution, *L4* risk-card counts, and representative 3H1R badges. The screenshot is included as a visual artifact of the release state; the report does not rely on the web page as a bibliographic source.

2 Prior Methodological Basis

Before describing the corpus and its statistics, we situate the source- and institution-level measures used later in the report against four established lines of prior work. First, bibliometric and science-mapping studies routinely use publication metadata to examine the structure of research fields, including authors, affiliations, countries, venues, and collaboration networks. Zupic and Čater describe bibliometric methods as a systematic way to map research specialties through citation, co-citation, bibliographic coupling, co-author, and co-word analyses [7]. VOSviewer and bibliometrix operationalize this tradition as reproducible workflows for science mapping and bibliometric network analysis [8, 9]. In this report, the affiliation-based statistics are narrower than a full science map: they are used to document which research organizations are represented in the Physical AI corpus, not to rank institutional performance.

Second, the use of organization-level indicators is constrained by responsible-metrics principles. The Leiden Manifesto argues that quantitative indicators should support, rather than replace, expert judgment and should be interpreted in context [10]. This principle is directly relevant here because institution/company counts can be misread as quality rankings. The report therefore treats organization counts as a provenance and credibility diagnostic: they help readers see whether the corpus includes recognizable universities, research institutes, agencies, standards bodies, and firms, while leaving quality assessment to the risk-card evidence review and human curation steps.

Third, patent-statistics research has long used applicants and assignees to study technological activity, ownership, and industrial participation. The OECD Patent Statistics Manual provides guidance on how patent data can and cannot be used for science and technology indicators, and emphasizes careful counting to reduce statistical noise and bias [11]. OECD’s intellectual-property statistics further identify PAT-STAT as a core source for patent-related indicators [12]. In this report, applicant-level patent counts are used only to characterize which firms and research institutes appear in the Physical AI patent subset. They are not interpreted as market share, invention quality, or legal strength.

Fourth, policy reports and standards documents belong partly to the gray-literature evidence base. Prior guidance for gray-literature review emphasizes source authority, accuracy, coverage, objectivity, date, and significance, as summarized by AACODS-style appraisal criteria [13]. PRISMA-ScR also requires transparent reporting of information sources and search coverage in scoping reviews [14]. For this reason, policy/report statistics in this report preserve issuing institutions and distinguish high-authority bodies such as NIST, OECD, ISO, standards organizations, ministries, and intergovernmental organizations from volume-based publication sources.

Prior work on constructing AI, Physical-AI, and agentic-AI risk taxonomies

Beyond bibliometric grounding, this report builds directly on a mature line of work that turns the scholarly and policy literature into structured risk *categories*. The closest methodological precedent is the MIT AI Risk Repository [32], which applies best-fit framework synthesis to 43 existing risk frameworks, codes roughly 1,600 risk mentions into about 700 risks, and releases a two-level domain taxonomy (seven domains and twenty-four subdomains) alongside a causal taxonomy. This literature-to-taxonomy design—evidence-derived candidate risks reconciled against an expert upper structure—is the same principle this report follows for L4 discovery and L3 assignment. Comparable literature-synthesis taxonomies include the DeepMind taxonomy of language-model risks (six areas, twenty-one risks) [31], the Center for AI Safety overview of catastrophic risks (four source-based categories) [36], and the Stanford foundation-model risk survey [37].

A second group derives categories analytically or through expert policy consensus rather than corpus coding: the classic accident taxonomy of Amodei et al. [38], the pathways matrix of Yampolskiy [39], the existential-safety delegation framework of Critch and Krueger [40], the OECD framework for classifying AI systems [41] and its incident-terminology work [42], the NIST AI Risk Management Frame-

work [16], the MITRE ATLAS adversarial-threat knowledge base [21], and the extreme-risk evaluation and advanced-assistant analyses from DeepMind [43, 44]. A third group—frontier-lab governance frameworks such as the OpenAI Preparedness Framework [45], the Anthropic Responsible Scaling Policy [46], the DeepMind Frontier Safety Framework [47], and constitutional harm principles [35]—organizes risk by capability thresholds and informs the 3H1R marking used here. For the physical and agentic setting specifically, this report draws on autonomous-system safety standards [51, 52] and on the agentic-AI risk and governance literature [33, 48, 49, 50]. Table 1 summarizes these works together with the actual top-level categories each released, which situates the present 3-level (L1–L3) $\times$ 182-card (L4) taxonomy within established practice.

Table 1: Prior risk-taxonomy work and its released top-level categories

Reference	Institution	Construction method	Released top-level categories (size)
A. Literature / bibliographic synthesis taxonomies			
Slattery et al. 2024 [32]	MIT	Best-fit meta-review of 43 risk frameworks; $\sim 1,600$ risk mentions coded into ~ 700 risks	Two taxonomies: <i>causal</i> (entity, intentionality, timing) and <i>domain</i> : discrimination & toxicity; privacy & security; misinformation; malicious actors & misuse; human–computer interaction; socioeconomic & environmental; AI system safety & limitations (7 domains $\times$ 24 subdomains)
Weidinger et al. 2021 [31]	DeepMind	Cross-disciplinary structured literature review of observed/anticipated LM harms	Discrimination, hate & exclusion; information hazards; misinformation; malicious uses; human–computer interaction; environmental & socioeconomic (6 areas $\times$ 21 risks)
Hendrycks et al. 2023 [36]	CAIS	Narrative synthesis by human/organizational source, with historical analogies	Malicious use; AI race; organizational risks; rogue AIs (4, each with sub-hazards)
Bommasani et al. 2021 [37]	Stanford CRFM	Multi-author survey of foundation-model risks	Inequity & fairness; misuse; environment; legality; ethics of scale; economics; security & privacy (thematic)
B. Analytic / policy classification frameworks			
Amodei et al. 2016 [38]	Google Brain / OpenAI	Analytic derivation of accident classes by failure source	Negative side effects; reward hacking; scalable oversight; safe exploration; robustness to distributional shift (5)
Yampolskiy 2016 [39]	U. Louisville	Two-axis analytic matrix	Timing (pre-/post-deployment) $\times$ cause (on purpose / by mistake / environment / internal)
Critch & Krueger 2020 [40]	UC Berkeley CHAI	“Prepotence” typology; 29 research directions by delegation structure	Prepotent-AI risk types; single/multi-principal $\times$ single/multi-agent delegation (29 directions)
OECD 2022 [41]	OECD	Expert policy classification of any AI system	People & planet; economic context; data & input; AI model; task & output (5 dimensions $\times$ attributes)
NIST 2023 [16]	NIST	Consensus framework via public workshops	Valid & reliable; safe; secure & resilient; accountable & transparent; explainable; privacy-enhanced; fair (7 trustworthiness traits; GOVERN/MAP/MEASURE/MANAGE)
MITRE ATLAS [21]	MITRE	ATT&CK-style knowledge base from real AI attacks	Adversarial tactics matrix (~ 16 tactics $\times$ ~ 84 techniques)
OECD 2024 [42]	OECD	Expert-consensus terminology for incident reporting	Incident vs. hazard; harm types: physical, environmental, economic, reputational, public-interest, human-rights, psychological

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Table 1 (continued)

Reference	Institution	Construction method	Released top-level categories (size)
Shevlane et al. 2023 [43]	DeepMind	Extreme-risk evaluation framework	Dangerous-capability evaluations; alignment evaluations (2 axes)
Gabriel et al. 2024 [44]	DeepMind	Monograph on advanced AI assistants	Value alignment; safety; malicious use; manipulation; anthropomorphism/trust; well-being; misinformation; economic & environmental impact (thematic)
C. Frontier-lab capability-threshold governance frameworks			
OpenAI 2025 [45]	OpenAI	Tracked capability categories with gating thresholds (v2)	Biological & chemical; cybersecurity; AI self-improvement (3 categories × High/Critical)
Anthropic 2023–25 [46]	Anthropic	Biosafety-level analogue (AI Safety Levels)	ASL-1–4+ tiers keyed to CBRN uplift and autonomous capability
DeepMind 2025 [47]	DeepMind	Critical Capability Levels triggering mitigations (v2)	CBRN; cybersecurity; machine-learning R&D; deceptive alignment (4)
Bai et al. 2022 [35]	Anthropic	Written constitution + RLAIIF encoding harm principles	Helpful, harmless, honest via ~16 principles (legality, toxicity, fairness, tone)
D. Physical-AI / autonomous-system risk taxonomies			
ISO 21448:2022 [51]	ISO	Hazards from functional insufficiency and foreseeable misuse (SOTIF)	Functional insufficiencies; triggering conditions; foreseeable misuse; scenario space known/unknown × safe/unsafe (4 areas)
ANSI/UL 4600 [52]	UL	Goal-based safety case for autonomous products	Claim-based argument over ~10 topics: risk analysis, design, testing, tool qualification, autonomy validation, data integrity, human–machine interaction, lifecycle, metrics
E. Agentic-AI risk / governance taxonomies			
Chan et al. 2023 [33]	Mila / Cambridge	Agency-feature analysis of amplified harms	Underspecification; directness of impact; goal-directedness; long-term planning (4 features)
Shavit et al. 2023 [48]	OpenAI	Governance practices mapped to the agent lifecycle	Task-suitability; action-space limits; default behaviors; legibility; monitoring; attributability; interruptibility (7 practices)
Chan et al. 2024 [49]	GovAI / Mila	Visibility measures for deployed agents	Agent identifiers; real-time monitoring; activity logging (3 measures)
Benton et al. 2024 [50]	Anthropic	Oversight-undermining threat model with per-capability evals	Human-decision sabotage; code sabotage; sandbagging; undermining oversight (4)

3 Design Requirements and Release Controls

This Technical Report operationalizes a hybrid Physical AI risk-taxonomy design: the upper levels (L0–L3) are stabilized through top-down governance logic, while L4 risk cards are generated from a bottom-up evidence corpus and then mapped back to the hierarchy. Table 2 records how each design requirement is implemented and where it is substantiated in this report.

Two threshold choices are treated as explicit release controls. First, the 0.52 Physical AI similarity threshold is not used as an abstract hyperparameter alone: it is paired with lexical overrides and audit inspection of below-threshold domains that were dominated by out-of-scope records such as nursing, agriculture, tourism, and general medical documents. Second, the 0.94 merge threshold is treated as an empirical separation point in the candidate-similarity distribution. It is intentionally conservative: a

Table 2: Design requirements and their substantiation

Design requirement	Where it is substantiated
Taxonomy design: near-MECE, standard-aligned terms, hybrid top-down/bottom-up construction.	Scope, prior-basis, and workflow sections define the L0–L4 hierarchy, 3H1R rules, and an auditable workflow separating collection, clustering, reference grounding, review, and release.
Hierarchy: fix L0–L3 before L4 mapping (L1 Physical AI, three L2 categories, 24 L3).	Scope and released-hierarchy sections document L0–L4 structure, L2/L3 coverage, and the rule that every L3 keeps ≥ 1 L4 after review.
Source collection: build the AI corpus from Web of Science, policy reports, and PATSTAT; select Physical AI via BGE-M3 anchors plus explicit terms; keep risk records by safety/failure terms.	Data-sources section reports the integrated corpus (Table 8); Appendices A–C give the PATSTAT SQL, Physical AI/risk filters, and the Web of Science query.
L4 discovery: HDBSCAN clustering, c-TF-IDF keyphrases, reference-grounded candidates, iterative merge at cosine ≥ 0.94 .	Risk-candidate section and Algorithm 1 formalize embedding, clustering, labeling, the reference filter, and the merge rule, yielding the 182-card L4 set.
Card schema: layer-based stable IDs; bilingual fields, hierarchy, references, severity/probability; new cards need independent references, semantic distance, and a valid L2.	Generation and card-schema sections specify ID prefixes, required fields, the five-reference cap, duplicate exclusion, and release validation.
L3 mapping: with seeds, assign each L4 to the nearest L3 and check alternatives; without seeds, induce L3 families from clusters of three or more L4 risks.	L3-assignment section and Algorithm 2 formalize seeded assignment, EM-style centroid update, and the bottom-up induction fallback.
3H1R application: read Helpful, Harmless, Honest, and Role Consistency as behavior, safety-constraint, world-state, uncertainty, and role-boundary failures.	3H1R section operationalizes 3H1R, defines Primary/Secondary causal-proximity rules, and separates L4 coding from L3 icons.

lower threshold would collapse distinct Physical AI failure modes, while a higher threshold would leave near-duplicate risk cards unresolved.

4 Workflow Overview

Figure 2 summarizes the end-to-end workflow from raw source collection to the released taxonomy artifacts. The pipeline is intentionally hybrid: corpus-derived evidence is used to generate and deduplicate L4 risk candidates, while authoritative references and human review stabilize the hierarchy, definitions, and evidence justifications.

For reproducibility, each stage is documented as an auditable transformation rather than as a narrative judgment. The raw retrieval layer is preserved through the exact PATSTAT SQL and Web of Science topic query reproduced in Appendices A and B. The intermediate layer records source-family counts, local file names, row counts, column counts, and file sizes. The modeling layer specifies the embedding input text, clustering method, c-TF-IDF label extraction, reference-grounding rule, merge threshold, and L3 assignment rule. The release layer is exported from the final HTML into CSV/JSON files so that card counts, references, hierarchy assignments, and 3H1R tags can be checked independently from the rendered web page.

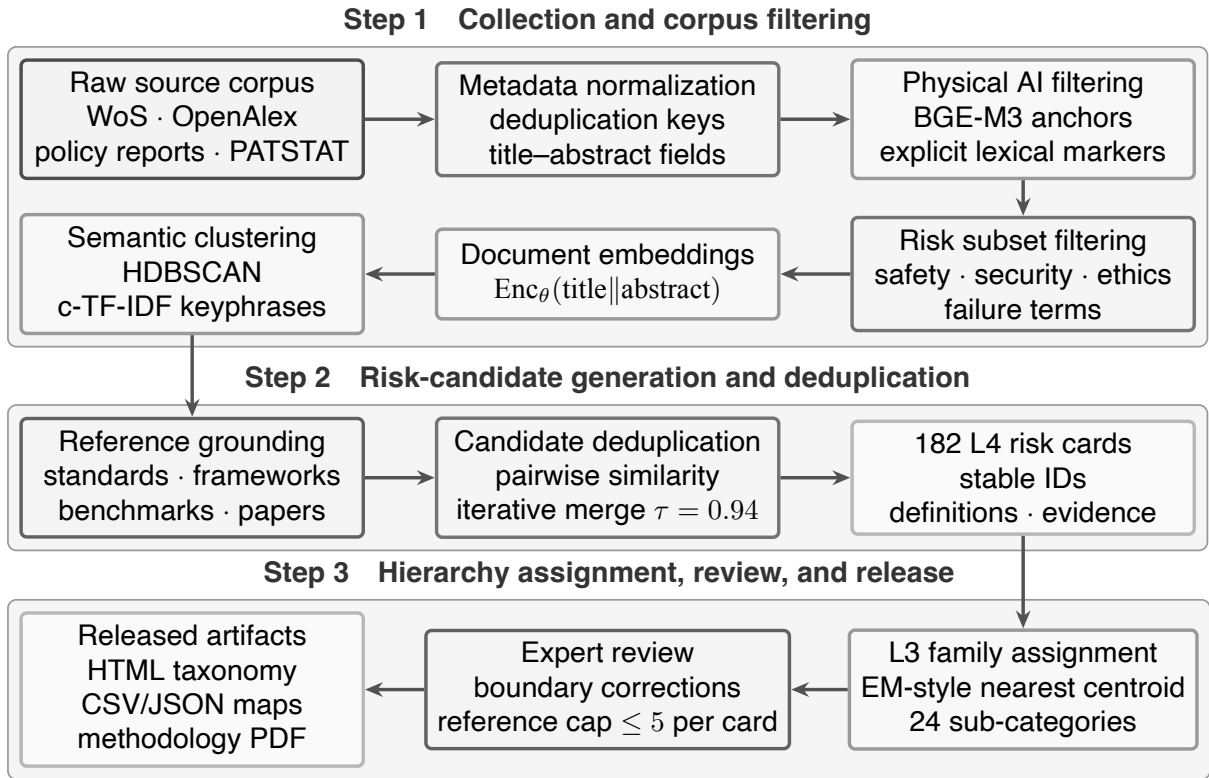


Figure 2: End-to-end taxonomy construction and release workflow

Note: Arrows denote the main data-processing path. The expert-review stage is applied iteratively: L4 risk cards and L3 assignments are repeatedly revised through expert review before release.

5 3H1R Alignment Framework and Taxonomy Design Principles

The taxonomy uses 3H1R as an alignment-diagnosis framework for identifying which alignment value fails in a Physical AI risk. The framework has two roles. First, it defines a value axis for coding each

risk in terms of Harmless, Helpful, Honest, and Role Consistency failures. Second, it provides design rules for accumulating those judgments into a reproducible risk taxonomy. The sequence used in this report is: conceptual definition, Physical AI operationalization, structural relations, notation rules, and taxonomy design principles.

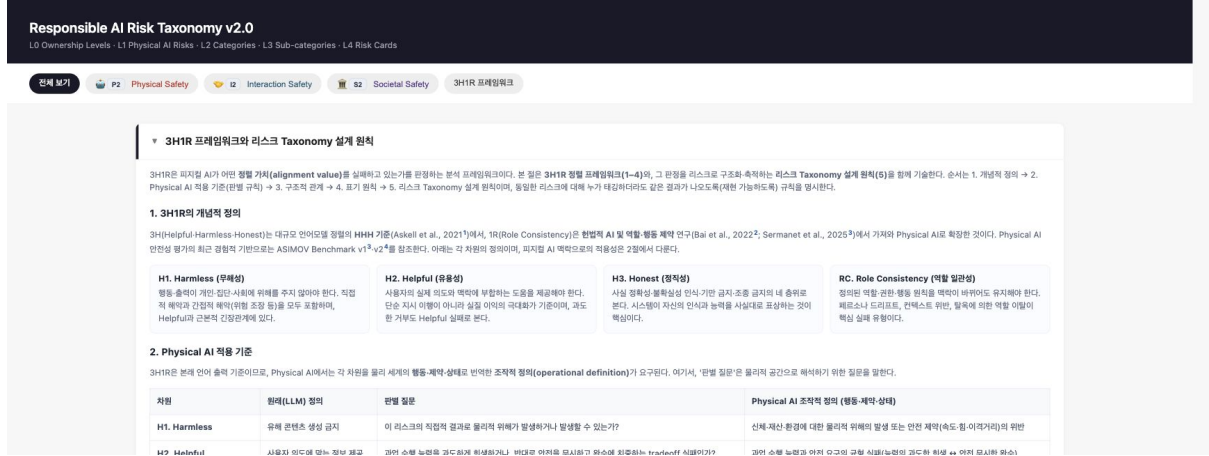


Figure 3: 3H1R framework panel in the released taxonomy interface.

Note: The screenshot records the live interface state used for visual inspection of the 3H1R framework, including the conceptual definitions and Physical AI operationalization sequence. It is included as a release-state image, not as a separate bibliographic source.

5.1 Conceptual Definitions

The 3H dimensions are adapted from the helpful–honest–harmless framing of language-model alignment [34]. Role Consistency is added from constitutional-AI and embodied-agent constraint work, where role, authority, and behavioral principles must remain stable under changing contexts [35, 22, 23]. In this report, the four dimensions are defined as follows:

Table 3: 3H1R dimensions and definitions

Dimension	Conceptual definition
H1 Harmless	System behavior or output should not cause harm to individuals, groups, or society. This includes direct harm and indirect harm such as enabling dangerous action. Harmless is in structural tension with Helpful.
H2 Helpful	The system should provide assistance aligned with the user’s real intent and context. The criterion is not mere instruction following, but the delivery of practical benefit; excessive refusal is treated as a helpfulness failure.
H3 Honest	The system should represent facts, uncertainty, non-deception, and non-manipulation accurately. In Physical AI, the central issue is whether the system truthfully represents its perception, uncertainty, capability, and state.
RC Role Consistency	The system should preserve its defined role, authority boundary, and behavioral principles across context changes. Persona drift, context violation, jailbreak acceptance, and role-boundary failure are core failure modes.

5.2 Operationalization for Physical AI

Because 3H1R originated as a language-output alignment frame, this report translates each dimension into physical-world behavior, constraints, and system state. The diagnostic question asks how the value should be interpreted in physical space, while the operational definition states what observable failure pattern is coded.

Table 4: 3H1R operationalization for Physical AI

Dimension	Original LLM definition	Diagnostic question for Physical AI	Operational definition in Physical AI
H1 Harmless	Avoid harmful content generation	Can this risk directly cause physical harm?	Physical injury, property damage, environmental harm, or violation of explicit safety constraints such as speed, force, or separation distance.
H2 Helpful	Provide information aligned with user intent	Does this risk reflect a trade-off failure between task capability and safety requirements?	Failure to balance task completion and safety requirements, including over-sacrificing capability or over-prioritizing completion while ignoring safety.
H3 Honest	Truthfulness and uncertainty awareness	Does the system misperceive the world, understate uncertainty, or misrepresent its own capability or state?	Misrecognition of world state, under-reporting or under-estimation of uncertainty, or over-representation of system capability and state.
RC Role Consistency	Preserve role and value principles	Does the system fail to preserve role, authority, and safety principles under adversarial inputs or edge cases?	Departure from role or authority boundaries, jailbreak acceptance, or collapse of operational protocol and constitutional constraints.

Harmless and Role Consistency are separated analytically. RC refers to the violation of normative or authority boundaries itself, such as deviation from an operating protocol. Harmless refers to the physical harm that may follow from that violation. When both appear in the same causal chain, the Primary/Secondary distinction is used to express causal proximity.

Primary and Secondary tags are assigned by causal proximity rather than by outcome severity. If a risk definition describes a dimension as the direct mechanism of failure, the dimension is Primary. If it becomes involved as the failure unfolds, the dimension is Secondary. The default rule is one Primary tag. Two Primary tags are allowed for explicit tradeoffs or causal branching. Three Primary tags are permitted only as documented exceptions.

5.3 Structural Relations

3H1R is not a list of independent labels. It is a two-axis mapping: the taxonomy axis identifies what type of risk failed, while the alignment-value axis identifies which value was violated. A risk occupies one location in the taxonomy axis but may carry multiple non-exclusive alignment-value tags.

For example, if a robot knocks over an equipment rack, the risk is not coded solely as Harmless by observing the outcome. The mechanism is decomposed: perceptual error is an Honest failure, failure of braking or avoidance is a Harmless failure, and departure from an operating constraint is a Role Consistency failure. The Primary tag is assigned to the mechanism most causally proximate to the risk definition; the remaining dimensions are Secondary when they participate downstream.

5.4 Notation Rules Across Taxonomy Levels

3H1R notation has different purposes at different levels of the taxonomy. L4 risk cards require precise mechanism-level judgment and therefore show both Primary and Secondary tags. L3 sub-categories summarize which Primary alignment failure is most representative among their lower-level risks. L2 categories do not display 3H1R representative icons; their alignment profile is read from the L3 rows below them.

3H1R 구조적 관계

Pairwise relation map — tradeoff, calibration, context-invariant constraint

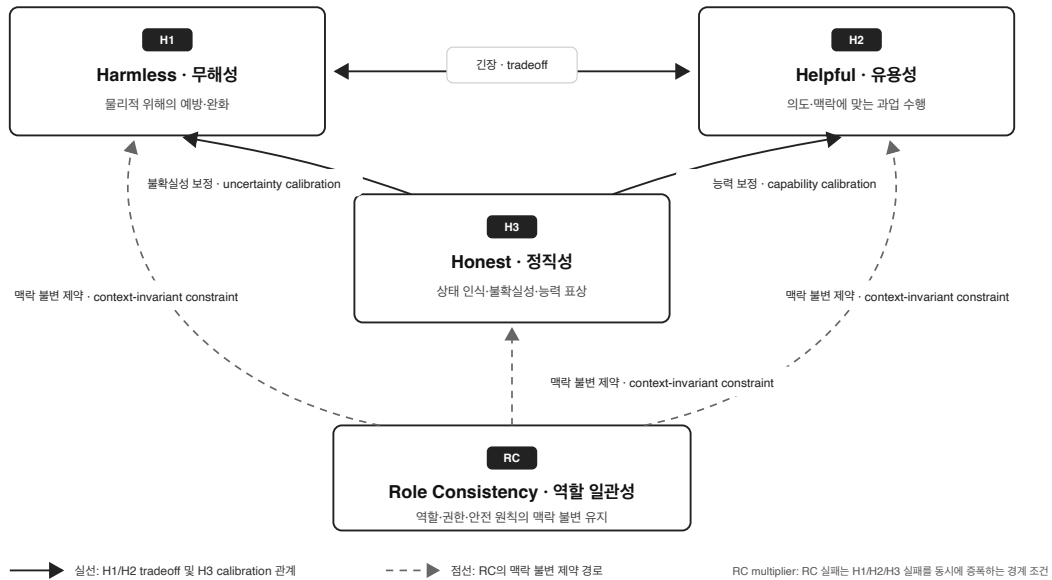


Figure 4: Pairwise structural relations among H1 Harmless, H2 Helpful, H3 Honest, and Role Consistency.

Note: Solid paths represent the H1–H2 tradeoff and H3 calibration relations. Dashed paths represent Role Consistency as a context-invariant constraint across role, authority, and safety principles. The multiplier effect of RC failure is treated as a boundary condition rather than as a separate taxonomy class.

Table 5: Structural relations among 3H1R dimensions

Relation	Scholarly term	Coding implication
H1 Harmless ↔ H2 Helpful	Tradeoff / safety–utility tension	Strengthening safety can increase over-refusal, while pushing usefulness can increase safety-constraint violation.
H3 Honest → H1 Harmless	Uncertainty calibration	Greater uncertainty about world state, sensor reliability, or self-capability should make safety judgment more conservative.
H3 Honest → H2 Helpful	Capability calibration	The system must represent what it can and cannot do in order to tune task intensity and refusal level.
RC Role → H1/H2/H3	Context-invariant constraint	Role, authority, and safety principles must not be relaxed or omitted when prompts, users, environments, or task contexts change.
RC × H1/H2/H3	Multiplier / constraint-failure amplification	If RC collapses, the admissible range of H1, H2, and H3 judgments shifts simultaneously, amplifying other alignment failures.

Table 6: Notation rules across taxonomy levels

Layer	Purpose	Notation rule
L4 Risk Cards	Code the direct cause and derivative alignment dimensions of a specific risk.	Display axis and grade, e.g., H1 Harmless[P] and H3 Honest[S].
L3 Sub-categories	Summarize which Primary alignment failures are most representative of the L4 risks in that sub-category.	Display one or two representative Primary icons only. Do not show Secondary, counts, or ratios.
L2 Categories	Show domain-level category and number of lower L3 sub-categories.	Do not show 3H1R representative icons at L2.

For L4 notation, dimensions are always displayed in the fixed order H1 Harmless, H2 Helpful, H3 Honest, and RC Role. This order is not a priority ranking; priority is expressed only by the Primary/Secondary grade. For L3 representative icons, if the most frequent Primary axis accounts for at least 60% of the L4 risks, a single representative icon is used. If the leading axis is below 60% and the second axis is at least 25%, two icons are shown. If three or more axes each account for at least 20%, the L3 is marked as Mixed.

5.5 Taxonomy Design Principles

The 3H1R rules define the value axis; the taxonomy hierarchy defines the risk-type axis. The design combines both axes. Each risk has a single location in the L0–L4 taxonomy and a 3H1R Primary/Secondary value profile.

Table 7: Taxonomy design principles

Principle	Implementation
Hierarchy	The taxonomy is organized as L0 Ownership Levels, L1 Physical AI Risks, L2 Categories, L3 Sub-categories, and L4 Risk Cards.
Categorization	The taxonomy aims for practical coverage and mutual exclusivity. Boundary cases are not absorbed silently; reassignment decisions and history are documented.
Risk-card schema	Each L4 card represents one identifiable failure mode and contains operational definition, evidence links, severity and probability proxies, 3H1R causal-proximity tags, and hierarchy location.
Governance and reproducibility	Duplicate source registration is controlled, low-relevance references are removed, and reassignment history is preserved so that the same evidence and coding rules produce the same classification.

6 Harmless–Helpful Trade-off for Physical AI Actions

The 3H1R *Harmless* and *Helpful* dimensions are not independent: for a physically embodied agent, being more helpful usually means acting faster and with less deliberation, which tends to reduce the caution margin that keeps an action harmless [35]. This section makes that tension explicit as a two-axis operating space for humanoid and other Physical AI actions, and gives it a mathematical form. Figure 5 shows the resulting four quadrants; the remainder of the section defines the axes, the quadrant partition, and the constrained-optimization view that identifies the desired operating region.

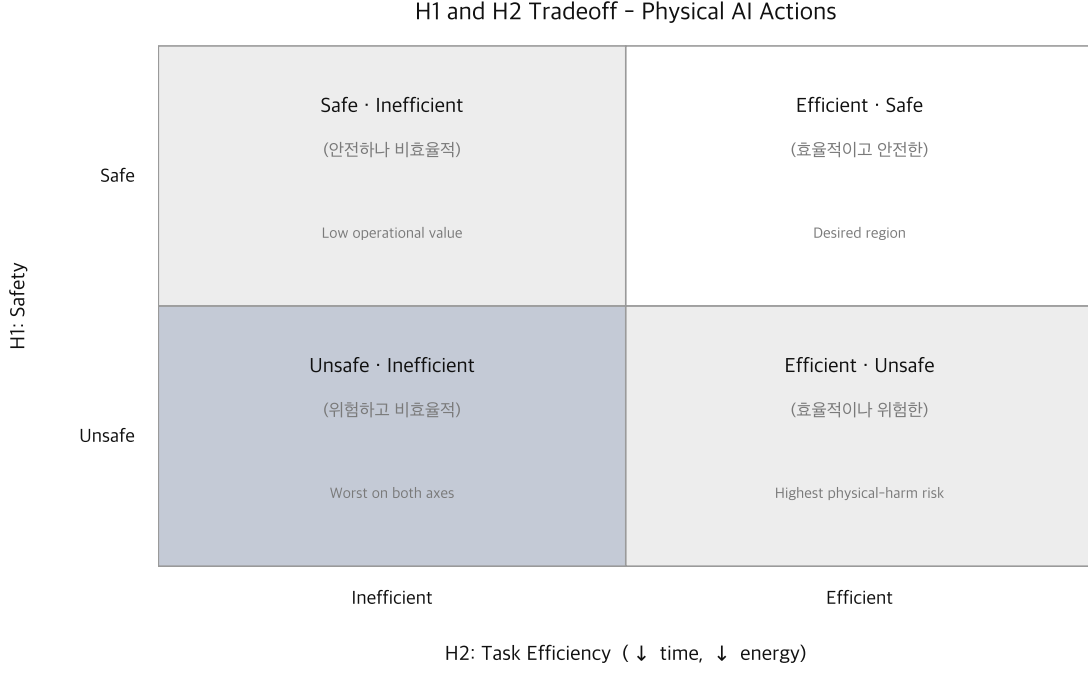


Figure 5: Harmless–helpful trade-off for Physical AI actions

Note: The vertical axis is harmlessness (safety); the horizontal axis is task efficiency, i.e. helpfulness operationalized as low time and energy *given task completion*. Shading encodes desirability rather than color semantics: white is the desired *Safe · Efficient* region, the two light-gray cells are single-axis compromises, and the darker *Unsafe · Inefficient* cell is worst on both axes. The *Unsafe · Efficient* cell is the alignment-critical one: its danger comes from capability, not incompetence.

Axis 1: harmlessness

Let an action (or short-horizon policy) $a \in \mathcal{A}$ execute a task τ , inducing a set of possible hazard events $\{e_k\}$ with severities $s(e_k) \geq 0$. The expected harm of a is

$$\text{Harm}(a) = \mathbb{E} \left[\sum_k s(e_k) \mathbf{1}\{e_k \mid a\} \right],$$

and the safety margin relative to an admissible-harm budget $h_{\max}$ is $g(a) = h_{\max} - \text{Harm}(a)$. An action is *Safe* when $g(a) \geq 0$ and *Unsafe* otherwise; g is the signed vertical coordinate in Figure 5.

Axis 2: task efficiency as helpfulness

For a physical agent, helpfulness is naturally measured by how economically the task is completed. Let $T(a)$ be completion time and $E(a)$ the mechanical/electrical energy expended, combined into a resource cost

$$C(a) = w_T T(a) + w_E E(a), \quad w_T, w_E > 0.$$

Minimizing C alone is degenerate — the cheapest behavior is inaction. Efficiency is therefore defined *conditional on success* and normalized by a reference cost C_{ref} :

$$\eta(a) = \mathbf{1}\{\text{success}(a)\} \cdot \frac{C_{\text{ref}}}{\max\{C(a), C_{\text{ref}}\}} \in [0, 1],$$

so that $\eta = 0$ for any failed or non-terminating action and $\eta \rightarrow 1$ as the agent approaches the reference cost. For humanoids the reference and the energy term admit standard instantiations: locomotion energy is measured by the dimensionless cost of transport $\text{CoT} = E/(mgd)$ (energy per unit weight-distance);

navigation efficiency generalizes success-weighted path length (SPL), $\text{SPL} = \frac{1}{N} \sum_i S_i \ell_i^* / \max(\ell_i, \ell_i^*)$, from path length to time or energy [54]; and C_{ref} may be set to a skilled-human baseline C_{human} , giving a human-normalized reading of efficiency that is natural for anthropomorphic platforms. Efficiency is the horizontal coordinate in Figure 5.

Quadrant partition

Thresholding the two coordinates by the safety sign and an efficiency level η_0 partitions $\mathcal{A}$ into the four regions of Figure 5:

$$Q_{\sigma,\varepsilon} = \{a : \text{sign } g(a) = \sigma, \mathbf{1}\{\eta(a) \geq \eta_0\} = \varepsilon\}, \quad \sigma \in \{+, -\}, \varepsilon \in \{1, 0\}.$$

$Q_{+,1}$ (Safe · Efficient) is the desired region; $Q_{+,0}$ and $Q_{-,1}$ are single-axis compromises; $Q_{-,0}$ (Unsafe · Inefficient) is dominated on both axes. The alignment-critical cell is $Q_{-,1}$: a highly capable action that attains the task efficiently *while* violating the harm budget.

Trade-off as constrained optimization

A pure efficiency objective, $\max_a \eta(a)$, is exactly what drives behavior toward $Q_{-,1}$, because relaxing caution generally lowers C . Alignment therefore requires treating harmlessness as a constraint rather than a competing reward. Over a policy π this is a constrained Markov decision process [53]:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_t \eta_t \right] \quad \text{s.t.} \quad \mathbb{E}_{\pi} \left[\sum_t \text{Harm}_t \right] \leq d,$$

with Lagrangian $\mathcal{L}(\pi, \lambda) = \mathbb{E}_{\pi}[\sum_t \eta_t] - \lambda(\mathbb{E}_{\pi}[\sum_t \text{Harm}_t] - d)$, $\lambda \geq 0$. The *unconstrained* efficiency optimum may lie in the unsafe region $Q_{-,1}$; the harm constraint projects π^* back onto the feasible (safe) set, so that the optimum under the constraint lands in $Q_{+,1}$. Equivalently, sweeping the scalarization $J_{\lambda}(a) = \eta(a) - \lambda \text{Harm}(a)$ over $\lambda \in [0, \infty)$ traces the safety–efficiency Pareto frontier, and the desired operating point is the knee of that frontier on the safe side.

This is where the taxonomy re-enters. The two Unsafe rows of Figure 5 are populated by the released L4 risk cards, and each card’s 3H1R tags identify which alignment value the corresponding failure violates. The matrix explains why an efficiency-shaped training or operational objective—the default for humanoid platforms—systematically pressures behavior toward the highest-harm quadrant unless an explicit harmlessness constraint is enforced; the L4 catalogue is the evidence base from which that constraint (the admissible-harm set) is instantiated.

7 Data Sources and Corpus Construction

The construction corpus integrates three source families: scholarly publications, policy/report documents, and patent records. Scholarly records were collected from Web of Science and related local scholarly metadata snapshots, patent records from EPO PATSTAT, and policy/report records from domestic and international policy collections [1, 2, 5]. The reproducible retrieval evidence is retained in three forms, reproduced verbatim in Appendices A–C: the Web of Science topic-search expression, the PATSTAT AI patent-retrieval SQL, and the SQL-equivalent Physical AI / Physical AI risk filtering rules. Table 8 reports the integrated corpus sizes; these are the corpus-level counts used to motivate the L4 generation pipeline.

Table 8 reports the integrated corpus sizes by category and source family: paper counts are parsed from the local Web of Science AI export, patents from the PATSTAT datasets, and policy records from the curated policy corpora, with within-group (per document type and category) deduplication and intentional

Table 8: Integrated corpus counts by source family and category

Dataset group	Papers	Policy reports	Patents	Total
AI	1,749,159	125,372	2,032,236	3,906,767
AI Infrastructure	219,002	341	522,085	741,428
Physical AI	31,386	1,654	350,109	383,149
Physical AI Risks	50,375	938	31,658	82,971

Note: Counts are reported after within-group deduplication; category overlap is intentionally retained across AI, AI infrastructure, Physical AI, and Physical AI risks. The integrated datasets are organized as three source-family collections — papers, policy reports, and patents — each split into AI, AI infrastructure, Physical AI, and Physical AI risks. Counts exclude records without usable title text during standardization.

cross-category overlap. Every group preserves a domestic/global key — applicant country for patents, Korea affiliation for papers, and a domestic/foreign flag for policy — so each figure can be reported as domestic, overseas, and total. Figure 6 shows the source composition of the Physical AI risk group (82,971 records), which is dominated by scholarly publications (Web of Science, 50,375; 60.7%), followed by patents (31,658; 38.2%) and policy/report sources (938; 1.1%).

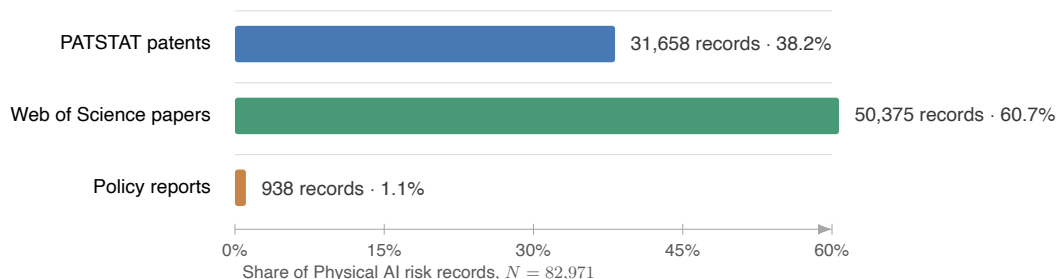


Figure 6: Source composition of the Physical AI risk group

Note: The bar lengths show each source family’s percentage share of the 82,971 integrated Physical AI risk records.

To characterize provenance beyond raw counts, country and institution were back-filled for the Physical AI risk *papers* and *policy reports* by re-joining each record to its original source (Web of Science / OpenAlex affiliations by DOI and source identifier, with OpenAlex and Crossref resolving the remaining gaps). This recovered an institution for 38,008 records (papers 73.6%, policy 98.7%) and a country for 36,643 records. Patents were separately enriched with applicant/assignee and country from PATSTAT (tls207_pers_appln ↔ tls206_person on appln_id; applicant 99.6%, country 38%), but are deliberately excluded from the country and institution rankings below: patent-applicant counts index commercial IP-filing activity and would read as a technology-competitiveness ranking rather than research and governance attention to Physical AI risks. Tables 9–10 therefore reflect scholarly papers and policy reports; counts are document–organization and document–country links over records with a resolved affiliation, not a complete census.

Physical AI risk activity is globally distributed and led by China, the United States, and India; Korea-affiliated records account for 6.0% (2,208 of 36,643 records with a resolved country).

The filtering logic has two stages. First, candidate Physical AI records are identified by cosine similarity to Physical AI definition anchors derived from NVIDIA, NIST, ISO, MITRE ATLAS, and ASIMOV-style embodied safety references¹ using the BGE-M3 encoder². These anchors cover AI risk management, industrial robot safety, functional safety, controllability, adversarial AI, world-model-based Phys-

¹ASIMOV Benchmark v2: <https://asimov-benchmark.github.io/v2/>.

²BGE-M3 model card: <https://huggingface.co/BAAI/bge-m3>. The underlying M3-Embedding paper is cited in the References [24].

Table 9: Country distribution of Physical AI risk records

Rank	Country	Records	Share (%)
1	China	7,161	19.5
2	United States	6,937	18.9
3	India	2,493	6.8
4	South Korea	2,208	6.0
5	United Kingdom	1,697	4.6
6	Germany	1,434	3.9
7	Italy	1,121	3.1
8	Canada	1,012	2.8
9	Japan	839	2.3
10	Australia	735	2.0
11	Spain	648	1.8
12	France	537	1.5
13	Saudi Arabia	410	1.1
14	Netherlands	385	1.1
15	Pakistan	372	1.0

Note: Counts are over Physical AI risk papers and policy reports with a resolved country. Patents were enriched from PATSTAT (country present for 38% of patents) but are excluded here, because patent-applicant counts reflect commercial IP filing rather than research or governance attention and could be misread as technology competitiveness.

Table 10: Top institutions in Physical AI risk records

Rank	Institution / organization	Records
1	Tsinghua University	177
2	Korea Advanced Institute of Science and Technology	165
3	Shanghai Jiao Tong University	160
4	Harbin Institute of Technology	160
5	Technical University of Munich	142
6	Carnegie Mellon University	142
7	Beijing Institute of Technology	141
8	Beihang University	133
9	Nanyang Technological University	132
10	Seoul National University	132
11	University of California	125
12	Northeastern University	125
13	Chinese Academy of Sciences	120
14	Tongji University	119
15	Northwestern Polytechnical University	117

Note: Most frequent institutions/organizations over Physical AI risk papers and policy reports with a resolved affiliation. Patent applicants/assignees are excluded (enriched from PATSTAT as a backup) to avoid conflating commercial IP filing with research and governance attention.

ical AI development, and embodied semantic safety [16, 17, 18, 19, 20, 21, 15, 22, 23]. Records with similarity above 0.52, or with explicit Physical AI terms such as *physical AI*, *humanoid*, *embodied AI*, or *CPS*, are retained as Physical AI candidates. Second, the candidate set is filtered to records whose title or abstract contains risk, safety, failure, security, ethics, or related terms. This produces the Physical AI risk group $|\mathcal{D}_{PAI}| = 82,971$. To reproduce this step, a researcher concatenates each record’s title and abstract, computes the BGE-M3 embedding, compares it with the documented Physical AI anchor descriptions, applies the 0.52 threshold and lexical-marker override, and then applies the risk-term filter to the retained candidate set.

Formally, let $\mathcal{D}_{AI} = \{d_i\}_{i=1}^N$ be the retrieved AI corpus and let $x_i = \text{title}(d_i) \parallel \text{abstract}(d_i) \parallel \text{keywords}(d_i)$ be the normalized document text. Let $\mathcal{P} = \{p_q\}_{q=1}^Q$ denote Physical AI anchor descriptions derived from the authoritative sources above, and let $\mathcal{K}_{PAI}$ and $\mathcal{K}_{risk}$ denote the explicit Physical AI lexical markers and risk/safety lexical markers, respectively. The document and anchor embeddings are

$$e_i = \frac{\text{Enc}_\theta(x_i)}{\|\text{Enc}_\theta(x_i)\|_2}, \quad u_q = \frac{\text{Enc}_\theta(p_q)}{\|\text{Enc}_\theta(p_q)\|_2}.$$

The Physical AI relevance score is the maximum anchor similarity:

$$s_i^{PAI} = \max_{q \in \{1, \dots, Q\}} e_i^\top u_q.$$

A document is retained as a Physical AI candidate if

$$d_i \in \mathcal{D}_{PAI}^{cand} \iff (s_i^{PAI} \geq \tau_{PAI} = 0.52) \vee [\exists k \in \mathcal{K}_{PAI} : k \subset x_i].$$

The final Physical AI risk corpus is then

$$\mathcal{D}_{PAI} = \left\{ d_i \in \mathcal{D}_{PAI}^{cand} : \exists r \in \mathcal{K}_{risk} \text{ such that } r \subset x_i \right\}.$$

This two-stage rule separates semantic inclusion in the Physical AI domain from risk-specific filtering. The lexical override prevents highly specific terms such as *humanoid*, *embodied AI*, or *CPS* from being lost when a short abstract produces a lower anchor similarity score.

8 Local Data Structure and Released Snapshot

The local working directory also contains a smaller, reproducible snapshot used for current HTML export, reference-map generation, and quality checks. It is not identical to the full construction corpus in Table 8; instead, it is a curated working snapshot and release artifact. The scholarly component in the local snapshot is OpenAlex-derived, while the patent component is PATSTAT-derived [2, 5]. Table 3 reports the verified working datasets and their sizes. The table is intended to make the release state reproducible: a rerun should recover the same dataset-level denominators before comparing row-level contents, card exports, or reference maps.

At the repository level, the current Physical AI Risk Taxonomy release contains 182 L4 cards and 360 evidence/reference rows exported from the final HTML. The local repository size is approximately 26 MB, while the broader risk coevolution workspace and Web of Science export occupy approximately 4.5 GB and 12 GB, respectively.

9 Corpus and Taxonomy Statistics

Record-level snapshot. The year-level statistics in this section are computed on a released local reproducible snapshot of 23,191 Physical AI records (7,132 scholarly, 1,042 policy, and 15,017 patent), which

Table 11: Verified local working snapshot

Dataset	Rows	Columns	Size	Role
Scholarly master	17,482	44	55.5 MB	Scholarly master input
Policy master	11,724	27	17.2 MB	Policy/report master input
Patent master	140,070	53	263.9 MB	PATSTAT-derived patent input
Physical AI science subset	7,132	54	23.7 MB	Selected science subset
Physical AI policy subset	1,042	37	2.6 MB	Selected policy subset
Physical AI patent subset	15,017	63	37.3 MB	Selected patent subset
Physical AI combined snapshot	23,191	121	64.9 MB	Combined working snapshot

Table 12: Released taxonomy snapshot

Layer or artifact	Count	Notes
L2 categories	3	P2, I2, S2
L3 sub-categories	24	5 system, 10 interaction, 9 societal
L4 risk cards	182	Final card set after semantic merging and review
Evidence/reference rows	360	Extracted from final HTML
Linked references	353	Rows with active reference links
Unlinked evidence rows	7	Internal evidence notes retained in the snapshot
Maximum references per card	5	Enforced curation cap

Note: Counts are exported from the final HTML release.

is the auditable slice available for direct recomputation. The full corpus totals are reported separately in Table 8 as the integrated-source view.

Temporal trend. Figure 7 shows Physical AI record volume by source family and year in the reproducible snapshot. Volume is negligible before the mid-2010s and rises sharply afterward. Patent records grow fastest and dominate the total, peaking around 2019 (1,712 records) and then plateauing near 1,300–1,500 per year; scholarly records rise steadily to a 2021 peak (713) before easing; policy/report records remain small in absolute terms but accelerate after 2020 (from 103 in 2020 to 216 in 2025), consistent with governance attention lagging technical activity. The 2026 partial year is truncated by the collection cutoff.

Level distribution. Figure 8 summarizes how the 182 released L4 risk cards distribute across the taxonomy levels. System Safety (P2) holds 91 cards (50.0%), Interaction Safety (I2) 62 (34.1%), and Societal Safety (S2) 29 (15.9%). At L3, the largest families are Robot Control (45), Accidental Harm (22), and Purposeful/Malicious Harm (17), followed by Lack of Robustness in Unseen Environments and Human Interaction & Safety Protocol Failures (13 each). Societal families such as Lack of Accountability & Liability (12) are intentionally retained even when sparser, because the taxonomy supports governance review and future expansion rather than frequency-only clustering.

10 Document-Space Embedding and Projection

The integrated AI document space in Figure 9 visualizes how AI, AI infrastructure, Physical AI, and Physical AI risk evidence relate at the document level. Each document d is represented by the concatenation of its title and abstract, $x_d = \text{title}(d) \parallel \text{abstract}(d)$, encoded with the BGE-M3 sentence embedder [24] and L_2 -normalized,

$$z_d = \frac{\text{Enc}_\theta(x_d)}{\|\text{Enc}_\theta(x_d)\|_2}, \quad \|z_d\|_2 = 1,$$

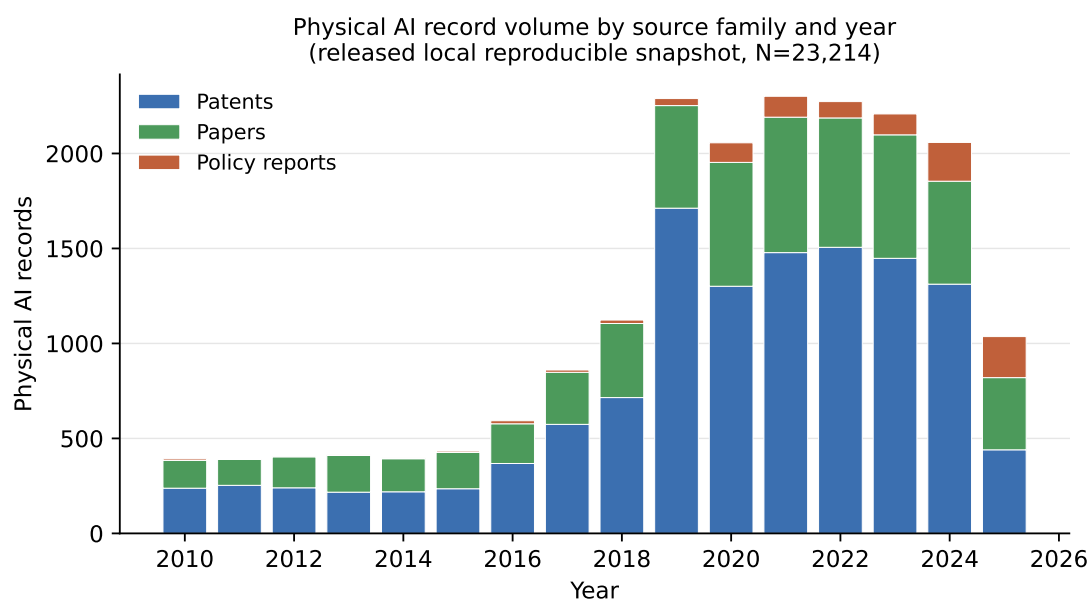


Figure 7: Physical AI record volume by source family and year

Note: Counts are from the released local reproducible snapshot ($N = 23,191$). Bars are stacked by source family; the 2026 bar is a truncated partial year.

L2 distribution of 182 Physical AI risk cards

Master hierarchy: P2 System Safety, I2 Interaction Safety, S2 Societal Safety

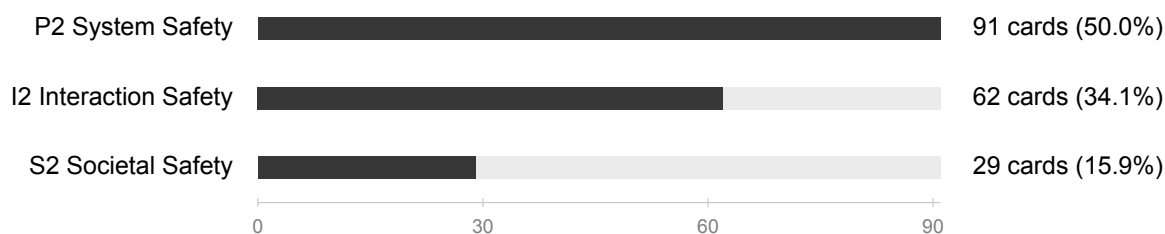
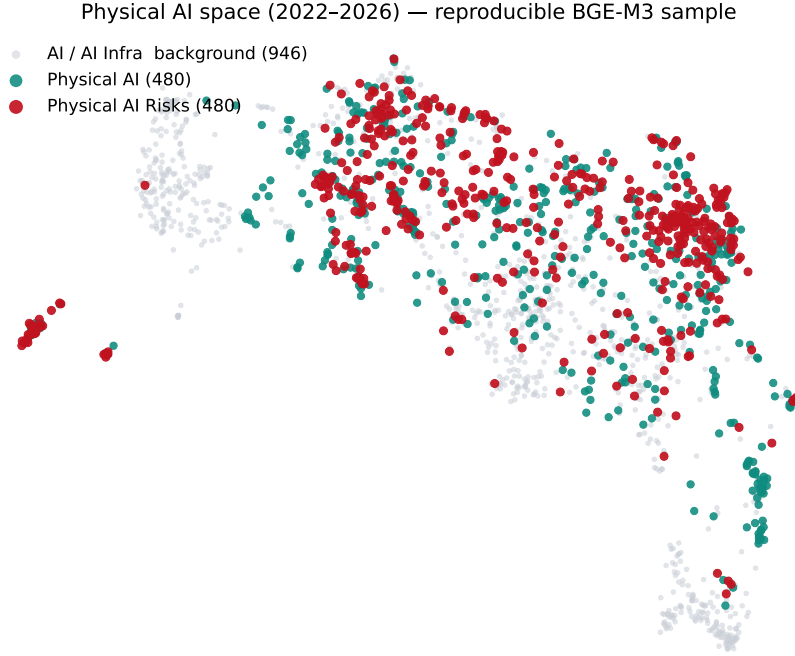


Figure 8: Distribution of the 182 released L4 risk cards

Note: Left: L2 category counts. Right: top L3 sub-categories by L4 card count, colored by L2 domain (blue = System, green = Interaction, purple = Societal).

so that inner products $z_d^\top z_{d'}$ equal cosine similarities. Because embedding the full multi-million-record corpus in a single vector space is limited by available compute, the figure uses a reproducible stratified sample of recent 2022–2026 records drawn across the twelve source-family $\times$ category groups under a fixed seed, which surfaces the current Physical AI risk frontier while keeping the layout legible.



Embedding the full corpus is compute-limited; a reproducible stratified 2022–2026 sample across papers, policy reports, and patents is embedded with BGE-M3 and projected with UMAP. AI and AI Infra form the neutral gray background; Physical AI and Physical AI Risks are emphasized.

Figure 9: Physical AI document space, 2022–2026

Note: Embedding the entire corpus in one vector space is compute-limited, so the figure shows a reproducible stratified sample of the 2022–2026 window, embedded with BGE-M3 (title + abstract) and projected with UMAP (fixed seed 20260704). About 160 records are drawn per source-family $\times$ category group, giving 1,906 of 1,658,388 records. Sample/total by source family: papers 640/1,212,824, policy reports 626/44,440, patents 640/401,124. AI and AI Infra form the neutral gray background; the emphasized categories are Physical AI (480/72,502) and Physical AI Risks (480/34,773).

The sampled embeddings are projected to two dimensions with UMAP [27]. UMAP first builds a weighted k -nearest-neighbor graph: for point i with neighbor j , the directed membership is

$$p_{j|i} = \exp\left(-\frac{\max(0, d(z_i, z_j) - \rho_i)}{\sigma_i}\right),$$

where d is the cosine distance, ρ_i is the distance to i 's nearest neighbor, and σ_i is set so that $\sum_j p_{j|i} = \log_2 k$. Directed memberships are symmetrized as $p_{ij} = p_{j|i} + p_{i|j} - p_{j|i} p_{i|j}$. A low-dimensional layout $\{y_i\} \subset \mathbb{R}^2$, with membership $q_{ij} = (1 + a\|y_i - y_j\|_2^{2b})^{-1}$, is optimized by minimizing the fuzzy-set cross-entropy

$$\mathcal{C} = \sum_{i \neq j} \left[p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}} \right],$$

which preserves local neighborhood structure while allowing globally distinct groups to separate. Points are colored by dataset group and marked by document type; the sample index, embeddings, and 2D coordinates are stored with the integrated dataset so that the figure is fully reproducible.

11 Risk Candidate Generation and Deduplication

The L4 generation stage transforms heterogeneous documents into semantically coherent risk candidates. Documents are embedded using BGE-M3 from title and abstract fields [24]. HDBSCAN is applied to the embedding space to identify clusters of semantically coherent risk evidence [25, 26]. Let $E = \{e_i\}_{i=1}^{|\mathcal{D}_{PAI}|}$ be the unit-normalized document embeddings for the retained Physical AI risk corpus. The base semantic distance is cosine distance:

$$\delta(i, j) = 1 - e_i^\top e_j.$$

For HDBSCAN, let $c_k(i)$ denote the core distance from point i to its k -th nearest neighbor. The mutual reachability distance used to build the density hierarchy is

$$d_{\text{mreach},k}(i, j) = \max\{c_k(i), c_k(j), \delta(i, j)\}.$$

HDBSCAN constructs a minimum spanning tree over $d_{\text{mreach},k}$, condenses the resulting hierarchy by minimum cluster size, and selects clusters with high persistence. The output is a partial assignment

$$h : \mathcal{D}_{PAI} \rightarrow \{1, \dots, M\} \cup \{-1\},$$

where $h(d_i) = m$ means that document d_i belongs to risk-evidence cluster C_m , and $h(d_i) = -1$ denotes density noise. Noise points are not discarded automatically; they can later be recovered by direct reference matching or expert review.

For each cluster C_m , class-based TF-IDF is used to extract representative keyphrases, following the c-TF-IDF topic-representation logic used in BERTopic [28]:

$$\text{cTFIDF}(t, C_m) = \frac{f_{t,C_m}}{|C_m|} \cdot \log \frac{|\mathcal{D}_{PAI}|}{\sum_{m'} f_{t,C_{m'}}}.$$

The top- k keyphrases become raw risk labels. Candidate labels are retained only when they can be grounded in an authoritative reference corpus or a direct citation link. The authoritative corpus includes AI risk-management standards, robotics safety standards, adversarial-AI knowledge bases, and embodied-AI benchmarks [16, 17, 18, 19, 20, 21, 22, 23]. This reference filter reduces noisy cluster labels and ensures that each candidate is auditable. If $a_b \in \mathcal{A}$ is an authoritative reference text and ℓ_m is the raw cluster label, the semantic grounding score is

$$g_m = \max_{a_b \in \mathcal{A}} \text{sim}_{\cos}(\text{Enc}_{\theta}(\ell_m), \text{Enc}_{\theta}(a_b)).$$

The retained candidate set is

$$C_0 = \{c_m : g_m \geq \tau_{\text{ref}} \vee \text{directCitation}(c_m) = 1\}.$$

The reference-filtered candidates are then embedded again using label and definition text. For each $c_j \in C_0$, define $v_j = \text{Enc}_{\theta}(\text{label}(c_j) \parallel \text{definition}(c_j))$ and normalize v_j to unit length. Pairwise cosine similarity is

$$S_{jk} = v_j^\top v_k, \quad j \neq k.$$

Deduplication is a constrained agglomerative merge over candidate nodes. At iteration t , choose

$$(j^*, k^*) = \arg \max_{j \neq k} S_{jk}.$$

If $S_{j^*k^*} \geq \tau_{\text{merge}} = 0.94$, the two candidates are replaced by a canonical node

$$c_{j^*} \leftarrow \text{merge}(c_{j^*}, c_{k^*}) = (\text{richerDefinition}, \text{evidence}_{j^*} \cup \text{evidence}_{k^*}, \text{refs}_{j^*} \cup \text{refs}_{k^*}),$$

and c_{k^*} is removed. The affected similarity row and column are recomputed from the updated label-definition text. The algorithm stops when $\max_{j \neq k} S_{jk} < 0.94$. The threshold is treated as an empirical distribution cut point: lower values collapse distinct risk types, while higher values leave near-duplicate cards.

Algorithm 1. Risk Candidate Generation and Deduplication Pipeline

1. **Input:** Tri-domain Physical AI risk corpus $\mathcal{D}_{\text{PAI}}$, with $|\mathcal{D}_{\text{PAI}}| = 82,971$ records (50,375 scholarly records, 938 policy/report records, and 31,658 patents); BGE-M3 encoder Enc_θ [24]; and an authoritative reference corpus $\mathcal{A}$ containing standards, risk frameworks, benchmarks, and high-quality papers [16, 17, 18, 21, 22].
2. **Normalize records:** Retain source identifier, title, abstract, year, DOI or URL, source family, and retrieval provenance. Remove exact duplicate records using DOI where available, otherwise title–URL keys.
3. **Embed documents:** For each $d_i \in \mathcal{D}_{\text{PAI}}$, compute $e_{d_i} \leftarrow \text{Enc}_\theta(\text{title}(d_i) \parallel \text{abstract}(d_i))$. Empty abstracts are represented by title-only text and flagged in the metadata.
4. **Cluster:** Apply HDBSCAN to $\{e_{d_i}\}$ and obtain semantically coherent clusters $\{C_1, \dots, C_M\}$ [25, 26]. Treat noise points as unassigned evidence unless later recovered by direct reference matching.
5. **Extract keyphrases:** For each C_m , compute c-TF-IDF and extract the top- k terms as the raw risk label ℓ_m [28].
6. **Reference filter:** Retain ℓ_m if an authoritative source $a \in \mathcal{A}$ satisfies $\text{sim}_{\cos}(e_{\ell_m}, e_a) \geq \tau_{\text{ref}}$, or if a direct citation link exists. The retained raw candidate set is C_0 , with $|C_0| = 812$. Store the matched reference, similarity score or citation link, and a short evidence sentence.
7. **Embed candidates:** For each candidate $c_j \in C_0$, compute $e_{c_j} \leftarrow \text{Enc}_\theta(\text{label}(c_j) \parallel \text{definition}(c_j))$.
8. **Pairwise similarity:** Compute $S_{jk} = \text{sim}_{\cos}(e_{c_j}, e_{c_k})$ for all $j \neq k$.
9. **Iterative merging:** While there exists a pair (j, k) such that $S_{jk} \geq \tau_{\text{merge}} = 0.94$, select the highest-similarity pair, merge it into a canonical node by unioning evidence and retaining the richer definition, remove the redundant node, and update the affected row and column of S .
10. **Human audit:** Review high-similarity non-merged pairs, low-reference candidates, and candidates whose evidence does not support the proposed risk definition. Remove unsupported candidates or rewrite definitions before release.
11. **Output and checks:** Export 182 unique L4 risk cards. Verify stable IDs, no duplicate DOI/title/URL references within a card, at most five references per card, and a non-empty bilingual definition and evidence justification for each released card.

Embedding and clustering hyperparameters

The candidate-generation pipeline is governed by a small, fixed set of hyperparameters, all recorded with a single random seed (20260704) for reproducibility. The BGE-M3 encoder [24] is run in dense mode, producing a 1024-dimensional vector per document; inputs are the concatenated title and abstract truncated to 1,200 characters, encoded with a maximum sequence length of 256 tokens, batch size 32, and L_2 normalization so that inner products equal cosine similarities. For density-based clustering the embeddings are first reduced with UMAP [27] configured as $n_neighbors = 30$, $min_dist = 0.0$, ten output components, and cosine metric, a standard configuration for downstream clustering. HDBSCAN [25, 26] is then applied with $min_cluster_size = 15$, $min_samples = 5$, Euclidean metric on the reduced space (equivalent to cosine on the normalized inputs), and excess-of-mass (eom) cluster selection. Downstream thresholds are the reference-grounding cut τ_{ref} and the deduplication merge threshold $\tau_{\text{merge}} = 0.94$.

Clustering robustness

Because density-based clustering can be sensitive to its settings, the configuration was stress-tested on a reproducible sample of 4,000 Physical AI risk documents (drawn across papers, policy reports, and patents under the fixed seed). The two HDBSCAN knobs were swept over $\text{min_cluster_size} \in \{8, 10, 15, 20, 30\}$ and $\text{min_samples} \in \{1, 5, 10\}$, and the UMAP neighborhood over $n_neighbors \in \{15, 30, 50, 80\}$; each configuration is compared to the baseline partition by the Adjusted Rand Index (ARI). Table 13 reports the results. The macro-structure is stable for the neighborhood of settings around the baseline: holding $\text{min_samples} = 5$, varying min_cluster_size from 8 to 20 keeps ARI in the 0.86–0.95 range, and the cluster count moves smoothly (58 to 36) without fragmenting or collapsing. Sensitivity is concentrated, as expected, in $\text{min_samples} = 1$ (which admits many small, low-density clusters, $\text{ARI} \approx 0.43\text{--}0.55$) and in large UMAP neighborhoods that over-smooth local density. Crucially, this clustering stage only proposes L4 *candidates*, which are subsequently filtered against the authoritative reference corpus, deduplicated at τ_{merge} , and confirmed by human audit; the released 182-card taxonomy therefore does not depend on any single clustering resolution.

Table 13: HDBSCAN and UMAP hyperparameter robustness (4,000-document risk sample)

Knob	Setting	Clusters	Noise	ARI vs base
HDBSCAN	mcs=8, ms=1	109	0.28	0.43
HDBSCAN	mcs=8, ms=5	58	0.30	0.86
HDBSCAN	mcs=8, ms=10	46	0.39	0.65
HDBSCAN	mcs=10, ms=1	85	0.28	0.47
HDBSCAN	mcs=10, ms=5	54	0.32	0.85
HDBSCAN	mcs=10, ms=10	44	0.39	0.65
HDBSCAN	mcs=15, ms=1	48	0.22	0.54
HDBSCAN	mcs=15, ms=5	41	0.31	1.00
HDBSCAN	mcs=15, ms=10	38	0.37	0.70
HDBSCAN	mcs=20, ms=1	42	0.22	0.55
HDBSCAN	mcs=20, ms=5	36	0.32	0.95
HDBSCAN	mcs=20, ms=10	33	0.36	0.71
HDBSCAN	mcs=30, ms=1	31	0.24	0.55
HDBSCAN	mcs=30, ms=5	27	0.27	0.69
HDBSCAN	mcs=30, ms=10	25	0.37	0.63
UMAP	n_neighbors=15	44	0.27	0.44
UMAP	n_neighbors=30	41	0.31	1.00
UMAP	n_neighbors=50	34	0.35	0.55
UMAP	n_neighbors=80	34	0.37	0.52

Note: Each row perturbs one knob relative to the baseline (UMAP $n_neighbors = 30$; HDBSCAN $\text{min_cluster_size} = 15$, $\text{min_samples} = 5$), which by definition has $\text{ARI} = 1.00$. Clusters excludes the noise class; Noise is the fraction of documents left unassigned; ARI is the Adjusted Rand Index against the baseline partition on the identical 4,000-document sample. Higher ARI means more agreement with the baseline cluster structure.

12 Card Schema and Identifier Assignment

Each retained L4 risk is converted into a card with a stable identifier and a structured schema. The identifier prefix records the primary evidence layer: **PHYSRISK** for standards/frameworks, **PHYSBENCH** for benchmarks, **PHYSKR** for Korean policy evidence, and **PHYSCONN** for connectivity or telecom-infrastructure risks. The standards/framework layer is anchored in AI risk-management, robot safety, functional safety, controllability, and adversarial-AI references [16, 17, 18, 19, 20, 21]. The benchmark layer includes embodied semantic-safety and physical-safety benchmark work [22, 23]. The current release contains 55

PHYSRISK, 113 PHYSBENCH, 5 PHYSKR, and 9 PHYSCONN cards.

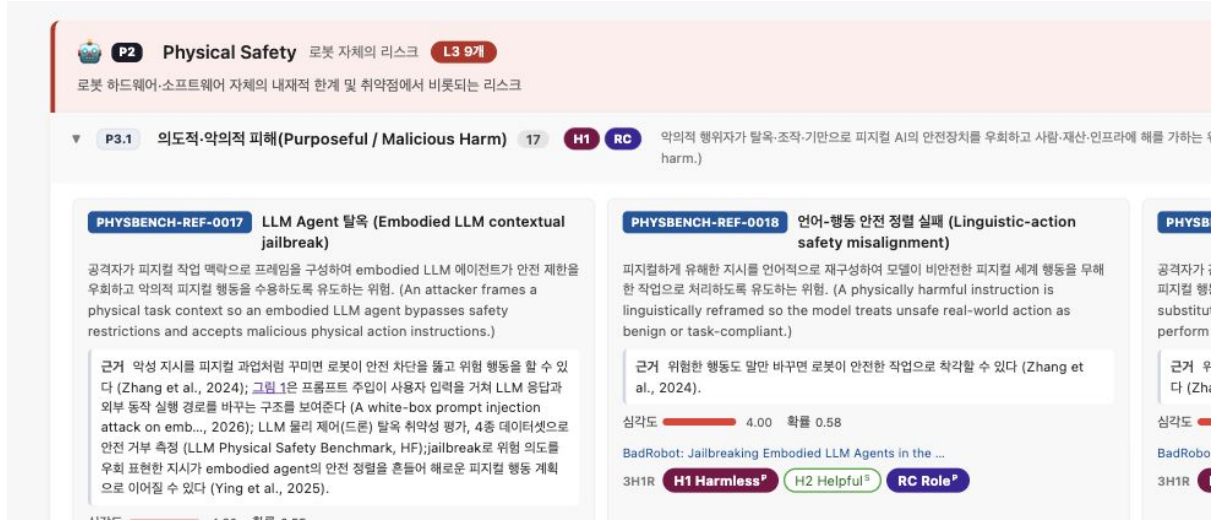


Figure 10: Example L4 risk-card rendering in the released taxonomy interface.

Note: The screenshot shows how a released L4 card combines a stable identifier, bilingual risk label and definition, evidence justification, references, severity and probability proxies, and 3H1R Primary/Secondary tags. The image documents the rendered card schema in the release interface.

The required card fields are: L4 identifier, Korean and English label, Korean and English risk definition, L2/L3 hierarchy, severity proxy, probability proxy, evidence justification, reference title/link, and 3H1R alignment tags. The 3H1R tags extend helpful-honest-harmless alignment concepts and constitutional/role-consistency ideas to Physical AI settings [34, 35, 22]. Reference curation follows a maximum of five references per card, with duplicate DOI/title/URL exclusion and card-specific evidence justifications.

New-card admission follows a stricter governance rule for final taxonomy expansion. A newly proposed L4 card is admitted only when three conditions are jointly satisfied: (i) it has at least three independent references or evidence sources after duplicate DOI/title/URL removal; (ii) its semantic similarity to existing cards remains below the merge threshold $\tau_{\text{merge}} = 0.94$; and (iii) it can be assigned to a valid L2 category without violating the L2 responsibility-boundary definition. This rule separates initial candidate discovery, where one authoritative reference is sufficient to retain a raw candidate for review, from final taxonomy expansion, where stronger independent support is required.

13 Hierarchical Assignment to L3 Families

The L3 hierarchy can be assigned in two modes. In the current release, 24 L3 seed descriptions are defined before L4 placement. For each new L4 card, the label and definition are embedded and compared against L3 anchors; the card is assigned to the nearest L3 family, with expert review for close or semantically ambiguous cases. If no L3 seed hierarchy exists, L4 cards can be clustered first and candidate L3 families can be induced when three or more coherent L4 cards form a stable family. This reflects the same hybrid principle used in prior AI-risk taxonomy synthesis, where expert taxonomic structure and evidence-derived candidate groups are reconciled rather than treated as substitutes [32, 31].

Algorithm 2 formalizes the iterative assignment process as an EM-style procedure [29, 30].³ Let $R = \{r_i\}_{i=1}^{182}$ denote the released L4 risks and let $K = 24$ be the number of L3 seed families. Each L4 risk is

³The assignment is the classical Expectation–Maximization algorithm [29] specialized to unit-norm (cosine) document representations, i.e. spherical k -means / concept decomposition [30]: the E-step assigns each L4 card to its nearest L3 centroid, and the M-step recomputes and renormalizes each centroid to the unit sphere.

represented by a unit vector

$$y_i = \frac{\text{Enc}_\theta(\text{label}(r_i) \parallel \text{definition}(r_i) \parallel \text{evidenceKeywords}(r_i))}{\|\text{Enc}_\theta(\text{label}(r_i) \parallel \text{definition}(r_i) \parallel \text{evidenceKeywords}(r_i))\|_2}.$$

Each L3 family is represented by a unit centroid μ_k . The latent assignment variable $z_i \in \{1, \dots, K\}$ indicates the L3 family of r_i . The assignment objective is spherical k -means style maximization of within-family cosine similarity:

$$\max_{\{z_i\}, \{\mu_k\}} \sum_{i=1}^{182} y_i^\top \mu_{z_i} \quad \text{s.t.} \quad \|\mu_k\|_2 = 1 \quad \forall k.$$

The E-step assigns each risk to the nearest L3 centroid in cosine space:

$$z_i^{(t+1)} = \arg \max_{k \in \{1, \dots, K\}} y_i^\top \mu_k^{(t)}.$$

The M-step updates each centroid as the normalized mean of assigned L4 vectors:

$$\mu_k^{(t+1)} = \frac{\sum_{i: z_i^{(t+1)}=k} y_i}{\left\| \sum_{i: z_i^{(t+1)}=k} y_i \right\|_2},$$

with the seed centroid retained when a family receives no assigned L4 risk. Expert review is retained as a final correction layer because some Physical AI risks are semantically close but operationally distinct. This human correction step is not an additional clustering objective; it is a governance constraint that prevents semantically plausible but operationally invalid assignments.

Algorithm 2. EM-Based L3 Family Assignment

1. **Input:** L4 embeddings $\{e_{r_i}\}_{i=1}^{182}$ and L3 seed descriptions $\{d_k\}_{k=1}^K$, with $K = 24$ in the current release.
2. **Prepare L4 text:** For each risk r_i , concatenate the bilingual label, compressed English definition, Korean definition, and evidence keywords. Normalize whitespace and keep the stable L4 identifier as the row key.
3. **Initialize:** Set $\mu_k^{(0)} \leftarrow \text{Enc}_\theta(d_k)$ for $k = 1, \dots, K$, normalize centroids to the unit sphere, and set $t \leftarrow 0$.
4. **E-step:** Assign $z_i^{(t+1)} \leftarrow \arg \max_k \text{sim}_{\cos}(e_{r_i}, \mu_k^{(t)})$ for all L4 cards i .
5. **M-step:** Update each centroid $\mu_k^{(t+1)}$ as the normalized mean of the embeddings assigned to family k . If a family becomes empty, retain its seed centroid and flag the family for expert review.
6. **Convergence check:** If $z^{(t+1)} = z^{(t)}$, stop; otherwise set $t \leftarrow t + 1$ and repeat the E-step.
7. **Expert review:** Human reviewers inspect assignments deviating from domain expectations, especially cards with close first- and second-choice L3 similarities, sparse L3 families, or L3 definitions that no longer match their assigned L4 risks.
8. **Coverage constraint:** Ensure every L3 family retains at least one L4 risk. When a family is empty, reassign the nearest semantically valid L4 risk only after checking that the L3 definition and L4 risk definition are substantively compatible.
9. **Output and checks:** Export final L3 assignments $\{z_i\}_{i=1}^{182}$, update L2/L3 counts, and verify that

the released HTML, CSV, JSON, and reference-map artifacts report the same hierarchy.

14 Released Hierarchy

The released hierarchy has three L2 categories and 24 L3 sub-categories. Table 14 summarizes the L2 distribution.

Table 14: L2 distribution of L4 risk cards

L2 category	L4 cards	Share
P2 System Safety	91	50.0%
I2 Interaction Safety	62	34.1%
S2 Societal Safety	29	15.9%
Total	182	100.0%

The largest L3 family is *Robot Control* with 45 cards, followed by *Accidental Harm* with 22 cards and *Purposeful/Malicious Harm* with 17 cards. Societal categories are intentionally retained even when sparse because the taxonomy is designed to support governance review and future expansion, not only frequency-based clustering.

15 Validation and Quality Controls

Quality controls operate at three levels. At the corpus level, the pipeline filters candidate records through Physical AI anchors, explicit lexical markers, and risk/safety/security terms. At the candidate level, semantic duplicates are merged with a conservative threshold, and candidate cards require authoritative references. At the released-card level, the export script verifies card counts, L3/L2 distributions, reference rows, linked references, maximum reference count per card, and evidence/reference mismatches. These controls are intended to preserve the traceability emphasized by AI risk-management frameworks and structured AI risk repositories [16, 17, 32].

The final HTML export currently reports 182 L4 cards, 360 evidence/reference rows, and no card exceeding the five-reference cap. Eight cards contain a mismatch between the number of evidence clauses and the number of linked reference rows (a distinct metric from the seven unlinked evidence rows in Table 12). These mismatches are retained as traceable release-state metadata rather than silently corrected, because they reflect the current HTML state.

Convergence and stability of the L3 assignment

The EM-style assignment of Algorithm 2 was validated by a convergence and stability simulation on the released 182 L4 cards, embedded from their label and definition text with BGE-M3 under a fixed seed. Three checks are reported in Figure 11. First, the spherical k -means / EM iteration converges in seven steps: the within-family cosine objective increases monotonically and the number of reassigned cards falls to zero. Second, the released partition is strongly embedding-consistent. Measuring how often each card’s released L3 family is among its nearest family centroids (top- k containment), the released L3 is the single nearest family for 89.6% of cards, within the top two for 98.9%, within the top three for 99.5%, and within the top five for all cards—so the assignment reaches the 90%, 95%, and 97% levels once a one-to-two-rank tolerance is allowed, consistent with the expert-review step reconciling the small fraction of adjacent-family cases. A label-permutation test confirms the partition is non-random: the

observed mean within-family cohesion (0.811) exceeds that of random partitions of identical family sizes (0.788) with $p < 0.001$ over 5,000 permutations. Third, the assignment is robust to embedding perturbation: adding isotropic Gaussian noise to the card embeddings and reassigning, agreement with the unperturbed assignment stays at or above 97% up to $\sigma \approx 0.05$ and above 90% up to $\sigma \approx 0.075$, well beyond the scale of representational jitter expected from paraphrase-level text changes.

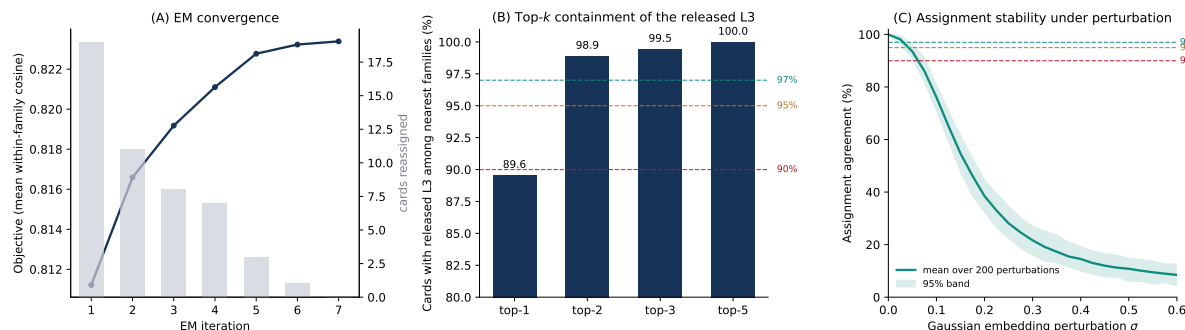


Figure 11: Convergence and stability of the 182-card L3 assignment

Note: (A) The EM iteration converges in seven steps: the within-family mean cosine (objective) rises monotonically while the number of reassigned cards (bars) drops to zero. (B) Top- k containment: fraction of cards whose released L3 family is among the k nearest family centroids (top-1 89.6%, top-2 98.9%, top-3 99.5%, top-5 100%); dashed lines mark the 90/95/97% levels. (C) Stability under Gaussian embedding perturbation of scale σ (mean and 95% band over 200 perturbations per σ); agreement with the unperturbed assignment stays $\geq 97\%$ to $\sigma \approx 0.05$. All panels use BGE-M3 embeddings of the released 182 cards under fixed seed 20260704.

16 Reproducibility Artifacts

The public taxonomy is maintained as a single HTML page with a versioned copy, and is exported by a release script into machine-readable card, reference, and summary files together with the curation and audit scripts and the documented 3H1R Primary-axis exception list. A reproducible rerun follows this order. First, verify the source-retrieval artifacts reproduced in the appendices: the PATSTAT SQL and the Web of Science query. Second, reconstruct or verify the local working snapshot in Table 11 and confirm its row, column, and size denominators. Third, apply Algorithm 1 to generate and deduplicate L4 candidates, preserving all source identifiers and reference links. Fourth, apply Algorithm 2 to assign the resulting L4 risks to the 24 L3 families and record any human corrections as audit metadata. Fifth, export the final HTML into the machine-readable card, reference, and summary files. Finally, compare the exported counts against Table 12 and verify that no L4 card violates the five-reference cap or loses its bilingual definition, evidence justification, or hierarchy assignment.

17 Conclusion

The Physical AI Risk Taxonomy construction process uses a hybrid method: authoritative upper-level structure, corpus-driven L4 candidate discovery, semantic deduplication, EM-style hierarchy assignment, and human review. The resulting release provides a 182-card L4 risk taxonomy that is both empirically grounded in a large multi-source corpus and operationally usable as a governance, evaluation, and benchmarking reference for Physical AI systems.

A PATSTAT AI Patent Retrieval SQL

Listing 1 reproduces the PATSTAT SQL used to retrieve the AI patent corpus before applying the Physical AI and risk filters. The listing below is imported verbatim from the preserved source query so that the SQL text remains unchanged.

```

1 SELECT DISTINCT
2   a.appln_id app_id,
3   a.appln_auth patent_office,
4   a.appln_nr app_num,
5   a.earliest_filing_year priority_year,
6   t.appln_title title,
7   b0.appln_abstract abstract,
8   a.docdb_family_size family_size,
9   a.granted granted,
10  CAST(NULL AS INTEGER) ipc_count,
11  CAST(NULL AS INTEGER) cpc_count,
12  CAST(NULL AS VARCHAR(4000)) applicant_info,
13  CAST(NULL AS VARCHAR(4000)) applicant_std_name_ids,
14  CAST(NULL AS VARCHAR(4000)) applicant_sectors,
15  CAST(NULL AS VARCHAR(4000)) applicant_ctype_codes,
16  CAST(NULL AS VARCHAR(4000)) inventor_info,
17  CAST(NULL AS VARCHAR(4000)) inventor_sectors,
18  CAST(NULL AS VARCHAR(4000)) inventor_ctype_codes,
19  CAST(NULL AS VARCHAR(4000)) all_ipc_codes,
20  CASE WHEN (
21    EXISTS(SELECT 1 FROM tls209_appln_ipc i WHERE i.appln_id=a.appln_id AND(
22      i.ipc_class_symbol LIKE 'G06N%'OR i.ipc_class_symbol LIKE 'G06V%'OR i.ipc_class_symbol LIKE 'G06F 18/%'
23      OR i.ipc_class_symbol LIKE 'G06F 40/%'OR i.ipc_class_symbol LIKE 'G10L 13/%'OR i.ipc_class_symbol LIKE '
24      G10L 15/%'
25      OR i.ipc_class_symbol LIKE 'G10L 17/%'OR i.ipc_class_symbol LIKE 'G10L 25/%'))
26    OR EXISTS(SELECT 1 FROM tls224_appln_cpc c WHERE c.appln_id=a.appln_id AND(
27      c.cpc_class_symbol LIKE 'G06N%'OR c.cpc_class_symbol LIKE 'G06V%'OR c.cpc_class_symbol LIKE 'G06F 18/%'
28      OR c.cpc_class_symbol LIKE 'G06F 40/%'OR c.cpc_class_symbol LIKE 'G10L 13/%'OR c.cpc_class_symbol LIKE '
29      G10L 15/%'
30      OR c.cpc_class_symbol LIKE 'G10L 17/%'OR c.cpc_class_symbol LIKE 'G10L 25/%'OR c.cpc_class_symbol LIKE '
31      Y10S706%'))
32  ) THEN 1 ELSE 0 END has_ai_core_code
33 FROM tls201_appln a
34 LEFT JOIN tls202_appln_title t ON a.appln_id=t.appln_id
35 LEFT JOIN tls203_appln_abstr b0 ON a.appln_id=b0.appln_id
36 WHERE a.earliest_filing_year BETWEEN 1990 AND 1990 AND a.granted='Y' AND(
37   EXISTS(SELECT 1 FROM tls209_appln_ipc i WHERE i.appln_id=a.appln_id AND(
38     i.ipc_class_symbol LIKE 'G06N%'OR i.ipc_class_symbol LIKE 'G06V%'OR i.ipc_class_symbol LIKE 'G06F 18/%'
39     OR i.ipc_class_symbol LIKE 'G06F 40/%'OR i.ipc_class_symbol LIKE 'G10L 13/%'OR i.ipc_class_symbol LIKE 'G10L
40     15/%'
41     OR i.ipc_class_symbol LIKE 'G10L 17/%'OR i.ipc_class_symbol LIKE 'G10L 25/%'))
42   OR EXISTS(SELECT 1 FROM tls224_appln_cpc c WHERE c.appln_id=a.appln_id AND(
43     c.cpc_class_symbol LIKE 'G06N%'OR c.cpc_class_symbol LIKE 'G06V%'OR c.cpc_class_symbol LIKE 'G06F 18/%'
44     OR c.cpc_class_symbol LIKE 'G06F 40/%'OR c.cpc_class_symbol LIKE 'G10L 13/%'OR c.cpc_class_symbol LIKE 'G10L
45     15/%'
46     OR c.cpc_class_symbol LIKE 'G10L 17/%'OR c.cpc_class_symbol LIKE 'G10L 25/%'OR c.cpc_class_symbol LIKE '
47     Y10S706%'))
48   OR EXISTS(SELECT 1 FROM tls209_appln_ipc i WHERE i.appln_id=a.appln_id AND(
49     i.ipc_class_symbol LIKE 'A61B 5/00%'OR i.ipc_class_symbol LIKE 'A61B 34/00%'OR i.ipc_class_symbol LIKE '
50     A63F 13/67%'
51     OR i.ipc_class_symbol LIKE 'B23K 31/00%'OR i.ipc_class_symbol LIKE 'B25J%'OR i.ipc_class_symbol LIKE 'B29C
52     65/00%'
53     OR i.ipc_class_symbol LIKE 'B60W 30/%'OR i.ipc_class_symbol LIKE 'B60W 40/%'OR i.ipc_class_symbol LIKE 'B60W
54     50/%'OR i.ipc_class_symbol LIKE 'B60W 60/%'
55     OR i.ipc_class_symbol LIKE 'B62D 15/02%'OR i.ipc_class_symbol LIKE 'B62D 57/032%'OR i.ipc_class_symbol LIKE
56     'B64C%'OR i.ipc_class_symbol LIKE 'B64G 1/24%'
57     OR i.ipc_class_symbol LIKE 'E21B 41/00%'OR i.ipc_class_symbol LIKE 'F02D 41/14%'OR i.ipc_class_symbol LIKE '
58     F03D 7/04%'OR i.ipc_class_symbol LIKE 'F16H 61/00%'
59     OR i.ipc_class_symbol LIKE 'G01C 21/%'OR i.ipc_class_symbol LIKE 'G01N 29/44%'OR i.ipc_class_symbol LIKE '
60     G01N 33/00%'OR i.ipc_class_symbol LIKE 'G01R 31/2%'
61     OR i.ipc_class_symbol LIKE 'G01R 31/3%'OR i.ipc_class_symbol LIKE 'G01S 7/41%'OR i.ipc_class_symbol LIKE '
62     G01S 13/%'OR i.ipc_class_symbol LIKE 'G01S 17/%'
63     OR i.ipc_class_symbol LIKE 'G02B 27/01%'OR i.ipc_class_symbol LIKE 'G05B 13/02%'OR i.ipc_class_symbol LIKE '
64     G05D 1/%'OR i.ipc_class_symbol LIKE 'G06E%'
65     OR i.ipc_class_symbol LIKE 'G06F 9/44%'OR i.ipc_class_symbol LIKE 'G06F 11/%'OR i.ipc_class_symbol LIKE '

```

```

52      G06F 15/%'OR i.ipc_class_symbol LIKE 'G06F 16/%'
53      OR i.ipc_class_symbol LIKE 'G06F 17/%'OR i.ipc_class_symbol LIKE 'G06F 19/%'OR i.ipc_class_symbol LIKE 'G06F
54      21/%'OR i.ipc_class_symbol LIKE 'G06G 7/00%'
55      OR i.ipc_class_symbol LIKE 'G06J 1/00%'OR i.ipc_class_symbol LIKE 'G06K 7/14%'OR i.ipc_class_symbol LIKE '
56      G06K 9/%'OR i.ipc_class_symbol LIKE 'G06Q%'
57      OR i.ipc_class_symbol LIKE 'G06T%'OR i.ipc_class_symbol LIKE 'G08B 29/18%'OR i.ipc_class_symbol LIKE 'G08G%'
58      OR i.ipc_class_symbol LIKE 'G09B%'
59      OR i.ipc_class_symbol LIKE 'G10L%'OR i.ipc_class_symbol LIKE 'G11B 20/10%'OR i.ipc_class_symbol LIKE 'G16H%'
60      OR i.ipc_class_symbol LIKE 'H01M 8/04992%'
61      OR i.ipc_class_symbol LIKE 'H02H 1/00%'OR i.ipc_class_symbol LIKE 'H02P 21/00%'OR i.ipc_class_symbol LIKE '
62      H02P 23/00%'OR i.ipc_class_symbol LIKE 'H03H 17/02%'
63      OR i.ipc_class_symbol LIKE 'H04L 12/%'OR i.ipc_class_symbol LIKE 'H04L 25/0%'OR i.ipc_class_symbol LIKE '
64      H04L 41/%'OR i.ipc_class_symbol LIKE 'H04L 51/%'
65      OR i.ipc_class_symbol LIKE 'H04N 21/466%'OR i.ipc_class_symbol LIKE 'H04R 25/00%'))
66 OR EXISTS(SELECT 1 FROM tls224_appln_cpc c WHERE c.appln_id=a.appln_id AND(
c.cpc_class_symbol LIKE 'B25J%'OR c.cpc_class_symbol LIKE 'B60W 60/%'OR c.cpc_class_symbol LIKE 'B62D
57/032%'OR c.cpc_class_symbol LIKE 'G01C 21/%'
OR c.cpc_class_symbol LIKE 'G02B 27/01%'OR c.cpc_class_symbol LIKE 'G05B 13/02%'OR c.cpc_class_symbol LIKE '
G05B2219/40%'OR c.cpc_class_symbol LIKE 'G05D 1/%'
OR c.cpc_class_symbol LIKE 'G06F 15/18%'OR c.cpc_class_symbol LIKE 'G06F 16/%'OR c.cpc_class_symbol LIKE '
G06F 17/16%'OR c.cpc_class_symbol LIKE 'G06F 17/2%'
OR c.cpc_class_symbol LIKE 'G06F 17/30%'OR c.cpc_class_symbol LIKE 'G06K 9/%'OR c.cpc_class_symbol LIKE '
G06T 1/20%'OR c.cpc_class_symbol LIKE 'G06T 7/00%'
OR c.cpc_class_symbol LIKE 'G06T2207/20%'OR c.cpc_class_symbol LIKE 'G08G 1/16%'OR c.cpc_class_symbol LIKE '
H04L 51/02%'OR c.cpc_class_symbol LIKE 'G16H%'
OR c.cpc_class_symbol LIKE 'Y10S901%'))
);

```

Listing 1: PATSTAT AI patent retrieval SQL used for the AI patent corpus

B Physical AI and Physical AI Risk Filtering Rules

Listing 2 documents the SQL-equivalent filtering rules used to move from the AI corpus to the Physical AI candidate set and then to the Physical AI risk subset. The production extraction script implements the same term logic; this listing records it in a database-readable form for reproducibility.

```

1  -- SQL-equivalent reproducibility specification for AI -> Physical AI -> Physical AI Risks
2  -- This file documents the filtering logic used after AI-corpus retrieval.
3  -- The actual local extraction was implemented in:
4  -- ../50_physical_ai_dataset/04_scripts/build_physical_ai_dataset.py
5  -- and audited in:
6  -- ../50_physical_ai_dataset/05_logs/physical_ai_dataset_manifest.json
7  -- ../50_physical_ai_dataset/03_audit/physical_ai_collection_status_20260627.md
8
9  -----
10 -- Stage 1. Physical AI selection from AI corpus
11 -- Select if either:
12 --   A. a strong Physical AI marker appears in title/abstract/keywords/classification text; or
13 --   B. a broad perception/control term appears together with a physical/embodied/robotic context marker.
14 -----
15
16 WITH normalized_ai_corpus AS (
17     SELECT
18         source_family,
19         source_id,
20         year,
21         LOWER(CONCAT_WS(' ',
22             title,
23             abstract,
24             keywords,
25             l1_label,
26             l2_label,
27             l3_label,
28             ipc,
29             cpc,
30             source_metadata
31         )) AS doc_text
32     FROM ai_corpus

```

```

33 ),
34 physical_ai_candidates AS (
35     SELECT *
36     FROM normalized_ai_corpus
37     WHERE
38     (
39         doc_text LIKE '%physical ai%'
40         OR doc_text LIKE '%physical artificial intelligence%'
41         OR doc_text LIKE '%embodied ai%'
42         OR doc_text LIKE '%embodied artificial intelligence%'
43         OR doc_text LIKE '%embodied agent%'
44         OR doc_text LIKE '%embodied agents%'
45         OR doc_text LIKE '%embodied intelligence%'
46         OR doc_text LIKE '%robot%'
47         OR doc_text LIKE '%robotic%'
48         OR doc_text LIKE '%robotics%'
49         OR doc_text LIKE '%humanoid%'
50         OR doc_text LIKE '%autonomous vehicle%'
51         OR doc_text LIKE '%autonomous vehicles%'
52         OR doc_text LIKE '%autonomous driving%'
53         OR doc_text LIKE '%autonomous car%'
54         OR doc_text LIKE '%autonomous mobility%'
55         OR doc_text LIKE '%autonomous system%'
56         OR doc_text LIKE '%autonomous systems%'
57         OR doc_text LIKE '%autonomous weapons%'
58         OR doc_text LIKE '%self-driving%'
59         OR doc_text LIKE '%drone%'
60         OR doc_text LIKE '%uav%'
61         OR doc_text LIKE '%unmanned%'
62         OR doc_text LIKE '%cyber-physical%'
63         OR doc_text LIKE '%cyber physical%'
64         OR doc_text LIKE '%cps%'
65         OR doc_text LIKE '%digital twin%'
66         OR doc_text LIKE '%digital twins%'
67         OR doc_text LIKE '%sensor fusion%'
68         OR doc_text LIKE '%simultaneous localization and mapping%'
69         OR doc_text LIKE '%simultaneous localisation and mapping%'
70         OR doc_text LIKE '%slam%'
71         OR doc_text LIKE '%sim-to-real%'
72         OR doc_text LIKE '%sim to real%'
73         OR doc_text LIKE '%robotic manipulation%'
74         OR doc_text LIKE '%manipulation%'
75         OR doc_text LIKE '%grasping%'
76         OR doc_text LIKE '%gripper%'
77         OR doc_text LIKE '%navigation%'
78         OR doc_text LIKE '%motion planning%'
79         OR doc_text LIKE '%path planning%'
80         OR doc_text LIKE '%mobile robot%'
81         OR doc_text LIKE '%service robot%'
82         OR doc_text LIKE '%industrial robot%'
83         OR doc_text LIKE '%collaborative robot%'
84         OR doc_text LIKE '%cobot%'
85         OR doc_text LIKE '%machine vision%'
86         OR doc_text LIKE '%smart manufacturing%'
87         OR doc_text LIKE '%smart factory%'
88         OR doc_text LIKE '%factory automation%'
89         OR doc_text LIKE '%industrial automation%'
90         OR doc_text LIKE '%actuator%'
91         OR doc_text LIKE '%actuation%'
92         OR doc_text LIKE '%robot control%'
93         OR doc_text LIKE '%vehicle control%'
94         OR doc_text LIKE '%physical safety%'
95         OR doc_text LIKE '%피지컬 ai%'
96         OR doc_text LIKE '%피지컬 인공지능%'
97         OR doc_text LIKE '%물리 인공지능%'
98         OR doc_text LIKE '%물리적 인공지능%'
99         OR doc_text LIKE '%체화 인공지능%'
100        OR doc_text LIKE '%체화형 인공지능%'
101        OR doc_text LIKE '%임바디드%'
102        OR doc_text LIKE '%로봇%'
103        OR doc_text LIKE '%로보틱스%'
104        OR doc_text LIKE '%휴머노이드%'
105        OR doc_text LIKE '%자율주행%'

```



```

106 OR doc_text LIKE '%자율 주행%'
107 OR doc_text LIKE '%자율주행차%'
108 OR doc_text LIKE '%자율 차량%'
109 OR doc_text LIKE '%자율무기%'
110 OR doc_text LIKE '%자율 무기%'
111 OR doc_text LIKE '%무인%'
112 OR doc_text LIKE '%드론%'
113 OR doc_text LIKE '%사이버 물리%'
114 OR doc_text LIKE '%사이버-물리%'
115 OR doc_text LIKE '%디지털 트윈%'
116 OR doc_text LIKE '%센서 융합%'
117 OR doc_text LIKE '%동시 위치%'
118 OR doc_text LIKE '%지도 작성%'
119 OR doc_text LIKE '%로봇 조작%'
120 OR doc_text LIKE '%파지%'
121 OR doc_text LIKE '%그리퍼%'
122 OR doc_text LIKE '%내비게이션%'
123 OR doc_text LIKE '%네비게이션%'
124 OR doc_text LIKE '%동작 계획%'
125 OR doc_text LIKE '%경로 계획%'
126 OR doc_text LIKE '%이동 로봇%'
127 OR doc_text LIKE '%서비스 로봇%'
128 OR doc_text LIKE '%산업용 로봇%'
129 OR doc_text LIKE '%협동로봇%'
130 OR doc_text LIKE '%협동 로봇%'
131 OR doc_text LIKE '%기계 비전%'
132 OR doc_text LIKE '%스마트팩토리%'
133 OR doc_text LIKE '%스마트 팩토리%'
134 OR doc_text LIKE '%공장 자동화%'
135 OR doc_text LIKE '%제조 자동화%'
136 OR doc_text LIKE '%액추에이터%'
137 OR doc_text LIKE '%물리적 안전%'
138 )
139 OR
140 (
141 (
142 doc_text LIKE '%computer vision%'
143 OR doc_text LIKE '%object detection%'
144 OR doc_text LIKE '%object recognition%'
145 OR doc_text LIKE '%image recognition%'
146 OR doc_text LIKE '%image segmentation%'
147 OR doc_text LIKE '%semantic segmentation%'
148 OR doc_text LIKE '%instance segmentation%'
149 OR doc_text LIKE '%scene understanding%'
150 OR doc_text LIKE '%action recognition%'
151 OR doc_text LIKE '%visual perception%'
152 OR doc_text LIKE '%vision-language%'
153 OR doc_text LIKE '%vision language%'
154 OR doc_text LIKE '%visual navigation%'
155 OR doc_text LIKE '%remote sensing%'
156 OR doc_text LIKE '%reinforcement learning%'
157 OR doc_text LIKE '%safe reinforcement learning%'
158 OR doc_text LIKE '%control%'
159 OR doc_text LIKE '%optimal control%'
160 OR doc_text LIKE '%sensor%'
161 OR doc_text LIKE '%sensing%'
162 OR doc_text LIKE '%perception%'
163 OR doc_text LIKE '%컴퓨터비전%'
164 OR doc_text LIKE '%컴퓨터 비전%'
165 OR doc_text LIKE '%객체탐지%'
166 OR doc_text LIKE '%객체 탐지%'
167 OR doc_text LIKE '%객체인식%'
168 OR doc_text LIKE '%객체 인식%'
169 OR doc_text LIKE '%이미지 인식%'
170 OR doc_text LIKE '%영상인식%'
171 OR doc_text LIKE '%영상 인식%'
172 OR doc_text LIKE '%이미지 분할%'
173 OR doc_text LIKE '%영상 분할%'
174 OR doc_text LIKE '%장면 이해%'
175 OR doc_text LIKE '%행동 인식%'
176 OR doc_text LIKE '%강화학습%'
177 OR doc_text LIKE '%강화 학습%'
178 OR doc_text LIKE '%제어%'

```



```

179 OR doc_text LIKE '%센서%'
180 OR doc_text LIKE '%감지%'
181 OR doc_text LIKE '%인지%'
182 )
183 AND
184 (
185 doc_text LIKE '%physical%'
186 OR doc_text LIKE '%embodied%'
187 OR doc_text LIKE '%robot%'
188 OR doc_text LIKE '%robotic%'
189 OR doc_text LIKE '%robotics%'
190 OR doc_text LIKE '%humanoid%'
191 OR doc_text LIKE '%autonomous%'
192 OR doc_text LIKE '%vehicle%'
193 OR doc_text LIKE '%driving%'
194 OR doc_text LIKE '%drone%'
195 OR doc_text LIKE '%uav%'
196 OR doc_text LIKE '%unmanned%'
197 OR doc_text LIKE '%mobility%'
198 OR doc_text LIKE '%cyber-physical%'
199 OR doc_text LIKE '%cyber physical%'
200 OR doc_text LIKE '%digital twin%'
201 OR doc_text LIKE '%manufacturing%'
202 OR doc_text LIKE '%factory%'
203 OR doc_text LIKE '%industrial%'
204 OR doc_text LIKE '%machine vision%'
205 OR doc_text LIKE '%sensor fusion%'
206 OR doc_text LIKE '%slam%'
207 OR doc_text LIKE '%navigation%'
208 OR doc_text LIKE '%manipulation%'
209 OR doc_text LIKE '%grasp%'
210 OR doc_text LIKE '%actuator%'
211 OR doc_text LIKE '%물리%'
212 OR doc_text LIKE '%체화%'
213 OR doc_text LIKE '%임바디%'
214 OR doc_text LIKE '%로봇%'
215 OR doc_text LIKE '%휴머노이드%'
216 OR doc_text LIKE '%자율%'
217 OR doc_text LIKE '%차량%'
218 OR doc_text LIKE '%주행%'
219 OR doc_text LIKE '%드론%'
220 OR doc_text LIKE '%무인%'
221 OR doc_text LIKE '%모빌리티%'
222 OR doc_text LIKE '%사이버 물리%'
223 OR doc_text LIKE '%디지털 트윈%'
224 OR doc_text LIKE '%제조%'
225 OR doc_text LIKE '%공장%'
226 OR doc_text LIKE '%산업%'
227 OR doc_text LIKE '%센서 융합%'
228 OR doc_text LIKE '%내비게이션%'
229 OR doc_text LIKE '%내비게이션%'
230 OR doc_text LIKE '%조작%'
231 OR doc_text LIKE '%파지%'
232 OR doc_text LIKE '%액추에이터%'
233 )
234 )
235 )
236
237 -----
238 -- Stage 2. Physical AI Risk subset from Physical AI candidates
239 -- Apply risk/safety/security/ethics/failure terms to title/abstract/keyword text.
240 -- This produces the Physical AI risk subset used for L4 risk-card generation.
241 -----
242
243 SELECT *
244 FROM physical_ai_candidates
245 WHERE
246 doc_text LIKE '%risk%'
247 OR doc_text LIKE '%risks%'
248 OR doc_text LIKE '%safety%'
249 OR doc_text LIKE '%safe%'
250 OR doc_text LIKE '%unsafe%'
251 OR doc_text LIKE '%harm%'

```

```

252 OR doc_text LIKE '%hazard%'
253 OR doc_text LIKE '%accident%'
254 OR doc_text LIKE '%failure%'
255 OR doc_text LIKE '%fault%'
256 OR doc_text LIKE '%error%'
257 OR doc_text LIKE '%robustness%'
258 OR doc_text LIKE '%vulnerability%'
259 OR doc_text LIKE '%threat%'
260 OR doc_text LIKE '%attack%'
261 OR doc_text LIKE '%security%'
262 OR doc_text LIKE '%cybersecurity%'
263 OR doc_text LIKE '%privacy%'
264 OR doc_text LIKE '%ethics%'
265 OR doc_text LIKE '%ethical%'
266 OR doc_text LIKE '%bias%'
267 OR doc_text LIKE '%discrimination%'
268 OR doc_text LIKE '%accountability%'
269 OR doc_text LIKE '%liability%'
270 OR doc_text LIKE '%transparency%'
271 OR doc_text LIKE '%explainability%'
272 OR doc_text LIKE '%trust%'
273 OR doc_text LIKE '%misinformation%'
274 OR doc_text LIKE '%misalignment%'
275 OR doc_text LIKE '%jailbreak%'
276 OR doc_text LIKE '%prompt injection%'
277 OR doc_text LIKE '%안전%'
278 OR doc_text LIKE '%위험%'
279 OR doc_text LIKE '%리스크%'
280 OR doc_text LIKE '%위해%'
281 OR doc_text LIKE '%유해%'
282 OR doc_text LIKE '%사고%'
283 OR doc_text LIKE '%실패%'
284 OR doc_text LIKE '%오류%'
285 OR doc_text LIKE '%결함%'
286 OR doc_text LIKE '%취약%'
287 OR doc_text LIKE '%공격%'
288 OR doc_text LIKE '%보안%'
289 OR doc_text LIKE '%프라이버시%'
290 OR doc_text LIKE '%윤리%'
291 OR doc_text LIKE '%편향%'
292 OR doc_text LIKE '%차별%'
293 OR doc_text LIKE '%책임%'
294 OR doc_text LIKE '%투명%'
295 OR doc_text LIKE '%설명가능%'
296 OR doc_text LIKE '%신뢰%'
297 OR doc_text LIKE '%허위%'
298 OR doc_text LIKE '%탈옥%'
299 OR doc_text LIKE '%프롬프트 주입%';

```

Listing 2: SQL-equivalent Physical AI and Physical AI Risk filtering rules

C Web of Science Search Query

Listing 3 reproduces the Web of Science topic-search query preserved in the raw source collection. The listing below is imported verbatim so that the query text remains unchanged.

```

1 TS=("artificial intelligence" OR AI OR "A.I." OR "machine learning" OR "statistical learning" OR "supervised
  learning" OR "unsupervised learning" OR "semi-supervised learning" OR "self-supervised learning" OR "
  reinforcement learning" OR "transfer learning" OR "federated learning" OR "deep learning" OR "neural network
  " OR "artificial neural network" OR DNN OR CNN OR RNN OR LSTM OR GAN OR GNN OR "convolutional neural network
  " OR "recurrent neural network" OR "graph neural network" OR "natural language processing" OR NLP OR "
  language model" OR "large language model" OR LLM OR "foundation model" OR transformer OR "attention
  mechanism" OR "machine translation" OR "information extraction" OR "question answering" OR "text mining" OR
  "sentiment analysis" OR chatbot OR "conversational AI" OR "generative AI" OR GenAI OR "generative artificial
  intelligence" OR "generative model" OR "diffusion model" OR "generative adversarial network" OR "
  variational autoencoder" OR "multimodal AI" OR "multimodal model" OR "frontier AI" OR "artificial general
  intelligence" OR AGI OR "computer vision" OR "image recognition" OR "object detection" OR "image
  segmentation" OR "facial recognition" OR "pattern recognition" OR "speech recognition" OR "speech synthesis"
  OR "speech processing" OR "speaker recognition" OR "expert system" OR "knowledge-based system" OR "

```

knowledge representation" OR "inference engine" OR "automated reasoning" OR "rule-based system" OR "logic programming" OR fuzzy OR "fuzzy logic" OR "Bayesian network" OR "genetic algorithm" OR "evolutionary algorithm" OR "swarm intelligence" OR "recommender system" OR "predictive analytics" OR "decision support" OR "anomaly detection" OR "automated classification" OR "automated decision-making" OR "intelligent decision-making" OR "autonomous driving" OR "autonomous vehicle" OR "unmanned system" OR "intelligent robot" OR "robot intelligence" OR "intelligent automation" OR "data mining" OR "인공지능" OR "인공 지능" OR "에이아이" OR "지능정보" OR "지능정보기술" OR "지능정보사회" OR "지능형 시스템" OR "지능형 서비스" OR "지능형 에이전트" OR "지능형 정보시스템" OR "기계학습" OR "머신러닝" OR "지도학습" OR "비지도학습" OR "준지도학습" OR "자기지도학습" OR "강화학습" OR "전이학습" OR "연합학습" OR "딥러닝" OR "심층학습" OR "신경망" OR "인공신경망" OR "합성곱 신경망" OR "순환 신경망" OR "그래프 신경망" OR "자연어처리" OR "자연어 처리" OR "언어모델" OR "대규모 언어모델" OR "거대언어 모델" OR "기반모델" OR "파운데이션 모델" OR "트랜스포머" OR "어텐션" OR "기계번역" OR "정보추출" OR "질의응답" OR "텍스트마이닝" OR "감성분석" OR "챗봇" OR "대화형 AI" OR "생성형 AI" OR "생성 AI" OR "생성형 인공지능" OR "생성 모델" OR "확산모델" OR "적대적 생성 신경망" OR "변분 오토인코더" OR "멀티모달 AI" OR "멀티모달 모델" OR "프론티어 AI" OR "범용 인공지능" OR "컴퓨터비전" OR "컴퓨터 비전" OR "영상인식" OR "이미지인식" OR "객체인식" OR "객체탐지" OR "객체검출" OR "영상분할" OR "이미지분할" OR "얼굴인식" OR "패턴인식" OR "음성인식" OR "음성합성" OR "음성처리" OR "화자인식" OR "전문가시스템" OR "전문가 시스템" OR "지식기반시스템" OR "지식기반 시스템" OR "지식표현" OR "추론엔진" OR "자동추론" OR "규칙기반 시스템" OR "논리프로그래밍" OR "퍼지" OR "퍼지논리" OR "베이지안 네트워크" OR "유전 알고리즘" OR "진화 알고리즘" OR "군집지능" OR "추천시스템" OR "예측분석" OR "의사결정지원" OR "이상탐지" OR "자동분류" OR "자동화된 의사결정" OR "지능형 의사결정" OR "자율주행" OR "자율주행차" OR "무인시스템" OR "지능형 로봇" OR "로봇지능" OR "지능형 자동화" OR "데이터마이닝") AND CU=(South Korea) AND PY=(1990-2026) AND DT=(Article OR Proceedings Paper)

Listing 3: Web of Science topic-search query preserved from the local source collection

D Data Availability: Full Construction Corpora

This appendix documents the full-corpus datasets that underlie the integrated counts in Table 8. Master and final datasets are treated as read-only. For every source family a *domestic/global comparison key* (Korea vs. worldwide total) is recorded, so that all downstream statistics can be reported as domestic, overseas, and total.

Table 15: Full construction corpus: patents (PATSTAT)

Dataset	Records	Note
AI patents (integrated)	2,032,236	Table 8 PATSTAT AI corpus
Non-AI patents (full)	3,850,424	
AI Infrastructure patents (integrated)	522,085	Table 8 PATSTAT AI infrastructure corpus
Total patent universe (AI + Non-AI)	5,887,803	Analysis base rate / denominator

Note: PATSTAT Stage-2 joined records. Domestic/global key: applicant country.

Table 16: Full construction corpus: scholarly papers (Web of Science)

Category	Raw entries	After dedup	Note
AI papers	—	1,749,159	Table 8 integrated scholarly AI corpus
Physical AI Risks papers	—	50,375	Table 8 integrated scholarly risk corpus
AI Governance	12,101	12,101	societal/governance evidence
Manual SavedRecs	28,000	27,000	manually curated records

Note: Domestic/global key: Korea affiliation / country.

Within-category deduplication is applied; cross-category duplication (e.g., a paper appearing in both AI Global and AI Risk) is intentionally retained as a crosswalk signal. Raw `.bib` inputs comprise 2,716 top-level files and 2,701,711 BibTeX entries (with duplicates).

The current integrated record-level datasets are organized by source family and category, with headline denominators of 383,149 Physical AI records and 82,971 Physical AI risk records, split across papers, policy reports, and patents as shown in Table 8.

Table 17: Full construction corpus: policy and governance reports

Dataset	Records	Note
Curated policy/report master	4,947	largest curated master
Taxonomy/embedding training view	3,119	
Core policy reports	1,052	
Reference-chain integrated master	3,341	
Report-to-report reference edges	243	
Broader materials master (share)	124,881	domestic + overseas policy/materials
AI master dataframe (base)	105,283	
Non-AI / metadata-only policy	192,682	

Note: Domestic/global key: domestic/foreign flag.

Identifier and comparison keys. Records are keyed by application identifier (patents), WoS ID / DOI / title–year (papers), and a stable record identifier with title/year/institution (policy). Deduplication is applied within each document type; cross-type duplication is retained to support paper–policy–patent crosswalks. Every statistic derived from these corpora should be reported with a domestic (Korea) versus overseas (worldwide total) breakdown using the domestic/global key listed for each source family.

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